
Towards Diffeomorphism-Equivariant Neural Networks via Canonicalization

Josephine Elisabeth Oettinger^{1 2} Zakhar Shumaylov² Johannes Bostelmann¹
Jan Lellmann¹ Carola-Bibiane Schönlieb²

Abstract

Incorporating group symmetries via equivariance into neural networks has emerged as a robust approach for improving efficiency and overcoming the data requirements of modern deep learning. While most existing approaches, such as group convolutions and averaging-based methods, focus on compact, finite, or low-dimensional groups with linear actions, this work explores how equivariance can be extended to infinite-dimensional groups. We propose a strategy designed to induce diffeomorphism equivariance in pre-trained neural networks via energy-based canonicalization. Formulating equivariance as an optimization problem allows us to access the rich toolbox of already established differentiable image registration methods. Empirical results on segmentation and classification tasks confirm that our approach achieves approximate equivariance and generalizes to unseen transformations without relying on extensive data augmentation or retraining.

1. Introduction

Deep learning models have become a cornerstone of modern data-driven research in numerous scientific fields (Abramson et al., 2024; Degraeve et al., 2022). In particular, in computer vision (Voulodimos et al., 2018) and medical imaging (Wang et al., 2020; Chen et al., 2022b), deep learning has become a central tool for tasks such as classification (Li et al., 2019), object detection (Zhao et al., 2019), and segmentation (Ronneberger et al., 2015). Despite this, these models’ reliance on large-scale, high-quality data and intensive computational requirements remains a major challenge (Goodfellow et al., 2016; Sun et al., 2017). Crucially,

¹University of Luebeck, Institute of Mathematics and Image Computing, Luebeck, Germany ²University of Cambridge, Department of Applied Mathematics and Theoretical Physics, Cambridge, United Kingdom. Correspondence to: Josephine Elisabeth Oettinger <josephine.oettinger@uni-luebeck.de>.

even with sufficient data, generalization properties are difficult to guarantee, as they depend heavily on the data seen during training and testing (Marcus, 2018; Zhang et al., 2017). In the quest for more robust and generalizable machine learning models, the focus has shifted towards exploiting known symmetries of the problem through *invariant* or *equivariant neural networks* (Bronstein et al., 2021). While *group invariance* means that a model’s output remains unchanged when the input is transformed by any element of the group, *group equivariance* means that the model’s output transforms in a predictable way under the group action that is applied to the input. More formally, let G be a group acting on differentiable manifolds $\mathcal{X}$ and $\mathcal{Y}$. A function $f : \mathcal{X} \rightarrow \mathcal{Y}$ is called *G -invariant* if for all $g \in G$ and $x \in \mathcal{X}$:

$$f(g \cdot_{\mathcal{X}} x) = f(x), \quad (1)$$

and *G -equivariant* if for all $g \in G$ and $x \in \mathcal{X}$:

$$f(g \cdot_{\mathcal{X}} x) = g \cdot_{\mathcal{Y}} f(x), \quad (2)$$

where the group element $g \in G$ acts on $\mathcal{X}$ and $\mathcal{Y}$ with corresponding actions $\cdot_{\mathcal{X}}$ and $\cdot_{\mathcal{Y}}$. For brevity, we will omit the subscripts when the action is clear from the context. The distinction between equivariance and invariance is illustrated in Figure 1, using segmentation and classification as representative tasks under the group of diffeomorphisms.

Enforcing invariance or equivariance with respect to a relevant transformation group can not only increase robustness but also reduce the need for extensive annotated data and therefore training time (Cohen & Welling, 2016; Bronstein et al., 2021; Bietti et al., 2021; Brehmer et al., 2025). Depending on the task, either group invariance (e.g., classification) or group equivariance (e.g., segmentation) is desired. Classically, the task is assumed to be symmetry-preserving with respect to the group considered; however, problems with approximate or local symmetries (Weijler & Hermosilla, 2025) as well as active symmetry breaking (Lawrence et al., 2025b) are also of interest.

Previous works have primarily focused on small, discrete symmetry groups, in particular rigid motions, or very specific learning settings (Cohen & Welling, 2016; Dieleman et al., 2015). However, in many applications, more generic

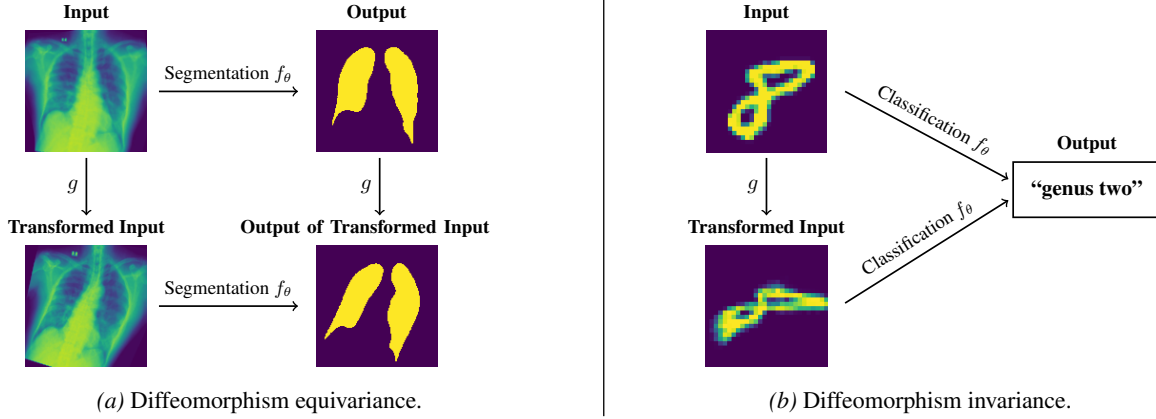


Figure 1. Examples of (a) diffeomorphism equivariance in lung image (Radiologi RSUA, 2023) segmentation and (b) diffeomorphism invariance in predictions of topological invariants in MNIST (LeCun et al., 1998). Applying a diffeomorphism $g \in \mathcal{D}(\mathcal{X})$ to the input should result in a correspondingly transformed segmentation output, i.e., $f_\theta(g \cdot x) = g \cdot f_\theta(x)$, and preserve the classification, i.e., $f_\theta(g \cdot x) = f_\theta(x)$. The classically grayscale color scheme is adjusted for better visualization.

nonlinear transformations are desirable. These transformations are naturally modeled as *diffeomorphisms*.

Definition 1.1 (Diffeomorphism). Let $\mathcal{X}$ and $\mathcal{Y}$ be differentiable manifolds. A map $g : \mathcal{X} \rightarrow \mathcal{Y}$ is called a *diffeomorphism* if it is bijective, differentiable, and has a differentiable inverse $g^{-1} : \mathcal{Y} \rightarrow \mathcal{X}$. The set of all such maps is denoted by $\mathcal{D}(\mathcal{X}, \mathcal{Y})$.

If $\mathcal{X} = \mathcal{Y}$, this set forms an infinite-dimensional group under composition, which we call $\mathcal{D}(\mathcal{X})$ (Omori, 1974). Diffeomorphisms are frequently used in imaging applications, particularly medical image registration (Younes, 2010; Dalca et al., 2019; Ashburner, 2007), where they model realistic anatomical transformations while preserving topological properties (Miller, 2004; Sotiras et al., 2013). A neural network that is equivariant to diffeomorphisms would therefore be highly valuable for real-world tasks, e.g., in cell segmentation in histopathology with variability in cell shapes; or lung segmentation with different breathing phases (Liu et al., 2025; Onieva Onieva et al., 2018).

To formalize our setting, let $\mathcal{X}$ denote the space of grayscale images, defined on the domain $\Omega \subseteq \mathbb{R}^2$ with intensities in $[0, 1]$, modeled as functions that map from Ω to $[0, 1]$, $\mathcal{X} := \{x \mid x : \Omega \rightarrow [0, 1]\}$. The group of diffeomorphisms $\mathcal{D}(\mathcal{X})$ acts on the image domain Ω . We define the group action of $\mathcal{D}$ on $\mathcal{X}$ as

$$(g \cdot x)(p) := (x \circ g)(p) = x(g(p)) \quad (3)$$

for all $p \in \Omega$, $x \in \mathcal{X}$ and $g \in \mathcal{D}(\mathcal{X})$. This group action produces deformed images, such as the input in Figure 2.

1.1. Contribution

In this work, we introduce Diffeomorphism-Equivariant Neural Networks (DiffeoNN)¹, a framework for adapting pre-trained models to achieve (approximate) diffeomorphism equivariance. Our approach extends the energy canonicalization methods of Shumaylov et al. (2024) and Kaba et al. (2023) from finite-dimensional Lie groups to the infinite-dimensional group of orientation-preserving diffeomorphisms. Unlike standard equivariance methods that require integration over locally compact groups, DiffeoNN utilizes group-agnostic canonicalization to achieve approximate equivariance without exhaustive sampling in practice. We parameterize these transformations using *stationary velocity fields* (SVFs), covering a broad class of complex spatial deformations, while being computationally feasible.

Extending energy canonicalization (Section 2) to the infinite-dimensional setting requires two technical modifications inspired by image registration (Younes, 2010). First, we propose a new energy functional (Section 3.2) that integrates a Variational Autoencoder (VAE) loss with an adversarial discriminator (Lunz et al., 2019; Mukherjee et al., 2024) and a deformation regularizer (Han et al., 2023; Bostelmann et al., 2026) to ensure structural consistency in the canonicalized outputs. Second, we replace the Lie-algebra optimization used in LieLAC (Shumaylov et al., 2024) with a gradient-based strategy specifically tailored for SVFs (Bostelmann et al., 2026). These adaptations provide a generalized framework applicable across various computer vision tasks, such as the segmentation example shown in Figure 2.

We evaluate DiffeoNN on approximate *diffeomorphism-equivariant segmentation* (Section 4.1) utilizing a syn-

¹Source code is available at <https://github.com/mic-uzl/DiffeoNN>

thetic dataset of nested squares and a real-world dataset of chest X-rays (Karmakar, 2024), as well as approximate *diffeomorphism-invariant homology classification* (Section 4.2), using the MNIST dataset (LeCun et al., 1998). Our results demonstrate that DiffeoNN consistently outperforms the naïve baseline and is on par with the data augmentation approach, while producing consistently fewer outliers. We further verify that the canonicalization step is in practice approximately diffeomorphism-invariant by construction and quantify the equivariance error.

1.2. Related Work

Unlike finite groups or simple transformations such as rotations and translations, the group of diffeomorphisms is infinite-dimensional. Consequently, it cannot be compactly parameterized for integration into standard equivariance approaches, like data augmentation (Shorten & Khoshgftaar, 2019), group-equivariant layers (Cohen & Welling, 2016), and averaging-based methods (Murphy et al., 2019; Puny et al., 2022). Given the limitations of these approaches, *canonicalization* emerges as the most viable strategy for achieving diffeomorphism equivariance. It is worth mentioning that while techniques from *Topological Deep Learning* can be used to construct topological invariants, achieving nontrivial geometric equivariance remains technically challenging, with most existing approaches focusing on restricted symmetry groups (e.g., Euclidean transformations) rather than infinite-dimensional diffeomorphism groups (Hajij et al., 2023; Ferrari et al., 2023; Eijkelboom et al., 2024).

Data Augmentation Data augmentation (Shorten & Khoshgftaar, 2019) offers a heuristic framework to approximate diffeomorphism equivariance (Hauberg et al., 2016) by expanding the training set through random group transformations. While conceptually viable, this approach encounters significant computational bottlenecks, primarily due to the necessity of sampling and transforming data within an infinite-dimensional space. As finite sampling cannot capture the full variability of the diffeomorphism group, this approach fails to provide rigorous guarantees of formal equivariance. Despite these limitations, it represents the primary alternative to canonicalization within existing literature; we therefore adopt it as our comparative baseline.

Group-Equivariant Layers In contrast to data-based methods, incorporating symmetries through equivariant architectures guarantees group equivariance by design. Convolutional neural networks (CNNs) are famously translation-equivariant due to weight sharing in their convolutional layers (Fukushima & Miyake, 1982; LeCun et al., 1989), claimed to be an important reason for their efficiency in imaging tasks (Krizhevsky et al., 2012; He et al., 2016).

This motivated *Group Convolutional Networks* (G-

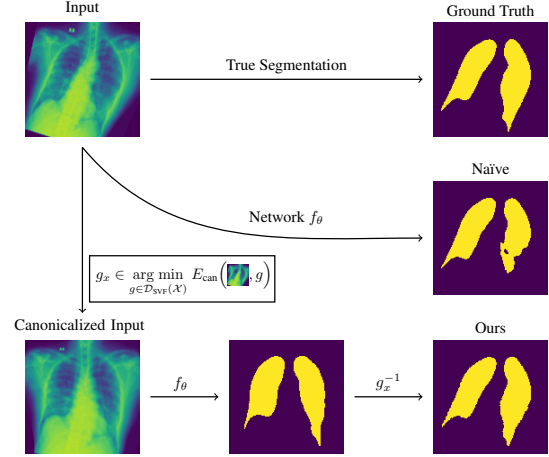


Figure 2. Canonicalization (DiffeoNN) for lung segmentation. The network f_θ is pre-trained on a simple chest X-ray/ground-truth segmentation dataset. Applying the trained model f_θ to a diffeomorphically transformed image without canonicalization results in an Intersection-over-Union (IoU) of 0.8769 (**naïve**) with clearly visible errors in the segmentation map. Using canonicalization, i.e., transforming the input closer to the training dataset before applying the network f_θ and reversing the transformation, achieves an almost perfect segmentation with an IoU of 0.9560 (**ours**).

CNNs) (Dieleman et al., 2015; Cohen & Welling, 2016), which replace standard convolutions with *group convolutions* over discrete symmetry groups, generalizing equivariance to arbitrary finite groups. G-CNNs have since been extended to compact groups such as $SO(2)$ and $SO(3)$ via spherical CNNs (Cohen et al., 2018), and to efficient implementations using fast Fourier transforms on groups (Kondor & Trivedi, 2018). While being a promising strategy for incorporating equivariance into neural networks, group convolutions require the group G to be finite (or at least compact and tractably parameterized (Cohen & Welling, 2016)), being implemented by a discrete sum over G , making G-CNNs incompatible with continuous, infinite, or non-representation groups such as the diffeomorphism group.

Lie group convolutions generalize to arbitrary Lie groups via integration with respect to the Haar measure with tractable implementations, e.g., different parameterizations (Finzi et al., 2020; Bekkers, 2021), exploiting the local Lie group structure (Kumar et al., 2024), and group-specific approaches for roto-translation groups (Bekkers et al., 2018). However, they too are inapplicable to our setting involving the group of diffeomorphisms, as they require the existence of a Haar measure (Milnor, 1984).

Averaging-based Approaches Averaging methods offer an alternative way to achieve equivariance without redesigning a neural network’s internal architecture. Instead of building specific symmetries into the layers, these methods oper-

ate on the *group orbit* of input x :

$$\mathcal{O}_x := \{g \cdot x \mid g \in G\}. \quad (4)$$

These strategies range from group averaging (Murphy et al., 2019), which integrates over the entire orbit, to frame averaging (Puny et al., 2022) and weighted frames (Dym et al., 2024), which utilize input-dependent subsets or measures to improve tractability. Despite these refinements, having to integrate over the group remains a bottleneck for diffeomorphisms, as the absence of a Haar measure makes defining a suitable integration measure computationally infeasible.

Canonicalization An approach closely related to, but distinct from frame averaging, is *orbit canonicalization* (Kaba et al., 2023). Canonicalization maps each input $x \in X$ to a *canonical representative* x_c within its orbit $\mathcal{O}_x$. By restricting the model’s domain to these representatives, the task complexity is reduced, allowing standard architectures to handle large or continuous groups. Canonicalization shifts complexity from averaging over an infinite-dimensional group to a non-trivial optimization problem. However, minimization requires only identifying a single element of the (infinite-dimensional) group rather than visiting all of them, which makes this approach highly efficient (Tahmasebi & Jegelka, 2025), robust to approximate symmetries (Kaba et al., 2023), with expressivity generalizable through probabilistic symmetry breaking (Lawrence et al., 2025a;b).

A notable extension of this paradigm is Lie Algebra Canonicalization (LieLAC) (Shumaylov et al., 2024), framing the *canonicalization step* as a preprocessing step. Crucially, it can be realized without explicit equivariant architectures, by defining the canonical representative as the minimizer of a task- and data-dependent energy function $E : X \rightarrow [0, \infty]$ over the group orbit:

$$c(x) := \arg \min_{y \in \mathcal{O}_x} E(y). \quad (5)$$

Intuitively, the energy E is chosen such that its minimizers are “close” to the training data X_E . LieLAC utilizes Lie algebra descent to find this minimizer, which, when combined with a pre-trained model trained on sparse data X_E , enables equivariance for arbitrary Lie groups without modifications of the underlying model. However, LieLAC is inherently restricted to finite-dimensional Lie groups, leaving the infinite-dimensional case as an open problem. In this work, we directly solve this by extending energy canonicalization to the problem of diffeomorphism equivariance.

2. Framework

Given a data manifold $\mathcal{X}$, the central idea of DiffeoNN is to train a simple model f_θ only on a sparse labeled training dataset $X_E \subseteq \mathcal{X}$ and without additional training, extend

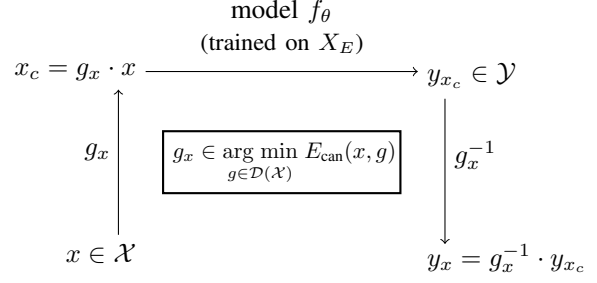


Figure 3. In the **canonicalization step**, an input $x \in \mathcal{X}$ is canonicalized to $x_c = g_x \cdot x$, where g_x is a canonicalization element that is determined by solving a task-specific optimization problem. Then the **inner model** f_θ is applied to x_c . Its output $y_{x_c} := f_\theta(x_c)$ is transformed by the **reverse canonicalization** g_x^{-1} to obtain the final output $y_x = g_x^{-1} \cdot y_{x_c}$.

it to perform well on the entire dataset $X \subseteq \mathcal{X}$ containing all diffeomorphic transformations of X_E . This allows us to make pre-trained networks equivariant without retraining.

Inspired by Shumaylov et al. (2024), our extended model $\tilde{f}_\theta$ consists of three steps: a *canonicalization step*, an *inner model trained on the training data X_E* , and a *reverse canonicalization step* (Figure 3):

1. In the **canonicalization step**, we find a representation x_c for a given input $x \in X$ such that x_c is “close” to the training data X_E . A *canonicalizing element* (diffeomorphism) $g_x \in \mathcal{D}(\mathcal{X})$ is found by minimizing a task-specific canonicalization energy:

$$g_x \in \arg \min_{g \in \mathcal{D}(\mathcal{X})} E_{\text{can}}(x, g). \quad (6)$$

The canonicalized input is then $x_c := g_x \cdot x$.

2. At the core is a **model** $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$, $x \mapsto f_\theta(x) =: y_x$ that is **trained on the smaller dataset X_E** . The range $\mathcal{Y}$ is a task-specific output space such as labels or segmentation masks. The model performs the task and outputs $y_{x_c} = f_\theta(x_c) \in \mathcal{Y}$ for the canonicalized input x_c .
3. To obtain the final output, a **reverse canonicalization** is applied:

$$\tilde{f}_\theta(x) := g_x^{-1} \cdot (f_\theta(g_x \cdot x)), \quad (7)$$

where $g_x^{-1} \in \mathcal{D}(\mathcal{Y})$ reverses the map corresponding to the canonicalizing element g_x on $\mathcal{Y}$. For example, if $\mathcal{X}$ and $\mathcal{Y}$ are both image spaces under the usual diffeomorphism action, g_x^{-1} would apply the inverse of g_x in order to obtain equivariance, while the trivial action in the $\mathcal{Y}$ space would result in invariance.

2.1. On Exact Equivariance

Exact Equivariance in Theory In a purely theoretical setup, canonicalization has been shown to result in an equiv-

ariant model (Kaba et al., 2023), i.e., $\tilde{f}_\theta(h \cdot x) = h \cdot \tilde{f}_\theta(x)$ for all $h \in G$ if the energy satisfies $E_{\text{can}}(h \cdot x, g) = E_{\text{can}}(x, g \cdot h)$ and the minimizer is unique. When uniqueness fails due to a nontrivial stabilizer subgroup, the construction of Kaba et al. (2023) requires “relaxed equivariance”. Shumaylov et al. (2024) and Ma et al. (2024) address this by considering the full frame $F_E(x) = \arg \min_{g \in G} E(g^{-1}x)$, which is strictly equivariant in a set-theoretic sense: for all g , $F_E(gx) = gF_E(x)$. Then, averaging (or any permutation-invariant aggregation) over the equivariant output set $\{gS_\theta(g^{-1}x) \mid g \in F_E(x)\}$ yields exact equivariance.

Approximate Equivariance in Practice In practice, for reasons of computational tractability and stability, we adopt three modifications that can break exact equivariance: we add regularization, breaking equivariance of E_{can} ; the group $\mathcal{D}(\mathcal{X})$ is finitely parameterized; and only a single approximate critical point of $E_{\text{can}}(x)$ is found in the minimization. Despite this, our energy functional in Section 3.2 still recovers approximate equivariance, as explained in Section 3.4.

3. Canonicalization for Diffeomorphisms

Applying the canonicalization framework to the group of diffeomorphisms $\mathcal{D}(\mathcal{X})$ requires defining the canonicalization step (5). In practice, this entails parameterizing $\mathcal{D}(\mathcal{X})$ and finding a suitable energy E_{can} that is small whenever $g \cdot x$ is “close” to the training data X_E , with g physically plausible, e.g., orientation-preserving, diffeomorphisms.

3.1. Parameterization via SVFs

In analogy with Dalca et al. (2019), we parameterize the group of diffeomorphisms $G = \mathcal{D}(\mathcal{X})$ using *stationary velocity fields* (SVFs). This yields $\mathcal{D}_{\text{SVF}}(\mathcal{X}) \subset \mathcal{D}(\mathcal{X})$, a structured subset of computationally efficient *SVF-generated diffeomorphisms*. Such transformations are commonly used in deformable image registration due to their smoothness, easily computable inverses, and well-defined theoretical structure (Younes, 2010; Ashburner, 2007; Dalca et al., 2019).

Definition 3.1 (Stationary Velocity Field (SVF)). A *stationary velocity field* is defined as a vector field $v : \Omega \rightarrow \mathbb{R}^d$, where $v(p)$ specifies the velocity at position p .

A velocity field v can be used to generate a *flow* φ :

Definition 3.2 (Flow). Given a stationary velocity field v , the associated *flow* $\varphi : \Omega \times [0, 1] \rightarrow \Omega$, $\varphi_t(x) := \varphi(x, t)$ is defined as the solution to the ordinary differential equation (ODE) with $t \in [0, 1]$ and $p \in \Omega$:

$$\frac{d\varphi_t(p)}{dt} = v(\varphi_t(p)), \quad \varphi_0(p) = p. \quad (8)$$

For $v \in \mathcal{C}_0^1(\Omega, \mathbb{R}^d)$, the solution of the ODE (8) at $t = 1$

yields a diffeomorphism $g := \varphi_1$ (Arguillere et al., 2014).

The solution map is defined by $\exp(v) := \varphi_1 = g$, called the *exponential map*, and connects the set of smooth vector fields to the group of diffeomorphisms (Younes, 2010; Trouné, 1998) similarly to the exponential map that connects a Lie algebra to its Lie group. Compared to the time-dependent, more general LDDMM formulation (Younes, 2010), SVF-generated diffeomorphisms are computationally more efficient, as the temporal integration can be performed by scaling and squaring (Arsigny et al., 2006a). While not every orientation-preserving diffeomorphism can be generated by an SVF (Hernandez et al., 2008), $\mathcal{D}_{\text{SVF}}$ suits our method as it includes $\text{id} := \exp(0)$, is closed under inverses, as $g^{-1} = \exp(v)^{-1} = \exp(-v)$, and it is expressive enough for our purpose; see also Appendix C.

3.2. Canonicalization Energy

The canonicalization energy E_{can} measures the similarity of a transformed image with the training data and encourages more “realistic” transformations within the allowed set of transformations $\mathcal{D}_{\text{SVF}}(\mathcal{X})$. In this work, the energy $E_{\text{can}} : X \times \mathcal{D}_{\text{SVF}}(\mathcal{X}) \rightarrow \mathbb{R}$ consists of an image similarity energy $E_{X_E} : X \rightarrow \mathbb{R}$ and a regularizer $E_{\text{reg}} : \mathcal{D}_{\text{SVF}}(\mathcal{X}) \rightarrow \mathbb{R}$:

$$E_{\text{can}}(x, g) := E_{X_E}(g \cdot x) + E_{\text{reg}}(g). \quad (9)$$

We again note that inclusion of a regularization term technically violates the conditions required for canonicalization to remain exactly equivariant (Kaba et al., 2023; Shumaylov et al., 2024). Nevertheless, the regularization term can be understood as a selection operator and approximate equivariance can be retained, see Section 3.4 and Appendix E.

Image Similarity Energy The image similarity energy estimates the similarity of an input to the training dataset X_E , defined for positive scalars λ_{VAE} and λ_{adv} as:

$$E_{X_E}(g \cdot x) := \lambda_{\text{VAE}} E_{\text{vae}}(g \cdot x) + \lambda_{\text{adv}} E_{\text{adv}}(g \cdot x), \quad (10)$$

The **VAE-based energy** E_{VAE} is the *variational autoencoder* (VAE) loss (Kingma & Welling, 2022), trained on the training dataset X_E . Low energy means the input can be well reconstructed by the VAE.

The **adversarial energy** E_{adv} is based on an adversarial discriminator trained to distinguish between the training images X_E (“real”) and diffeomorphically transformed ones (“fake”) following (Lunz et al., 2019; Mukherjee et al., 2024; Shumaylov et al., 2024). The “fake” diffeomorphically transformed images are generated by randomly sampling deformation parameters from a radial basis function parameterization (Forti & Rozza, 2014) to warp the training images X_E . E_{adv} is then the adversarial energy of the trained discriminator, and is smaller for inputs x that are “closer” to the

training dataset X_E . Crucially, training this discriminator is both stable and efficient, requiring minimal overhead.

Regularizing Energy In order to encourage physically plausible transformations, we follow (Han et al., 2023; Bostelmann et al., 2026) and define the **regularizer** E_{reg} as

$$E_{\text{reg}}(g) := \underbrace{\lambda_{\Delta} \|\nabla v\|_2^2}_{\text{gradient loss}} + \underbrace{\lambda_{\mathcal{J}} \|\max(0, -\det(\mathcal{J}_g))\|_1}_{\text{Jacobian determinant loss}}, \quad (11)$$

where λ_{Δ} and $\lambda_{\mathcal{J}}$ are positive scalar weights, v the SVF that induces g , and $\det(\mathcal{J}_g)$ the Jacobian determinant of g . The gradient loss is large for deformations with high spatial variation in their flow field. Therefore, it is small when the deformation is smooth. The Jacobian determinant loss penalizes negative Jacobian determinants. The intent is to enforce $\det(\mathcal{J}_g(p)) > 0$ for all $p \in \Omega$, as such diffeomorphisms are orientation-preserving, ensuring physical plausibility. A negative determinant indicates local reversal of orientation, such as folding or tearing, which is undesirable in most image processing tasks. Therefore, restricting to diffeomorphisms with a positive Jacobian determinant is a common practice in image processing and has been widely researched (Liu et al., 2023; Kuang, 2019).

Note that the canonicalization energy can be adjusted per task if desired. However, our approach is already highly versatile, as it only depends on training data and the group of diffeomorphisms, see Section 4. For more details and an ablation study regarding the energy terms, see Appendix D.

3.3. Canonicalization via Gradient-based Optimization

It remains to solve the canonicalization step (6). In contrast to the approach of LieLAC (Shumaylov et al., 2024), which uses Lie algebra descent, we use a gradient-based optimization method similar to (Dalca et al., 2019; Balakrishnan et al., 2019) to minimize the canonicalization energy E_{can} . We follow (Han et al., 2023; Bostelmann et al., 2026) in using a *Sinusoidal Representation Network (SIREN)* (Sitzmann et al., 2020) to parameterize the SVFs and a *Scaling and Squaring* approach (Arsigny et al., 2006a) to compute the solution $g_{\theta} = \exp(v_{\theta})$ of the ODE.

In practice, we can only expect the canonicalization step to be *approximately* diffeomorphism-invariant: Except for very rare cases, exact minimizers are numerically inaccessible due to the finite iteration budget of the gradient-based optimization scheme (Bostelmann et al., 2026).

3.4. Recovery of Approximate Equivariance

The unfortunate drawback of making canonicalization practically feasible is that exact equivariance may be violated, as explained in Section 2.1. However, the ways in which it is broken retain approximate equivariance.

On Regularization of the Canonicalization The full frame is often too large, as the whole stabilizer subgroup is contained inside: $g \text{Stab}_G(x) \subseteq F_E(x)$ for $g \in F_E(x)$. Yet, they are redundant for averaging over, as quotienting preserves equivariance:

$$g(F_E(x)/\text{Stab}_G(x)) = F_E(gx)/\text{Stab}_G(gx). \quad (12)$$

The regularization therefore simply acts as a selection operator (e.g., minimal-norm), common in inverse problems (Benning & Burger, 2018), and bypasses the continuity constraints of canonicalization (Dym et al., 2024), formalized in Appendix E, Theorem E.5.

On Parameterization of the Group Unlike classical equivariance methods, such as group convolutions, in canonicalization, the group parameterization only enters the optimization step. It naturally tolerates coarse parameterizations, or those lacking closure under composition, provided it is rich enough for the optimization step to generate canonicalized elements that are sufficiently close to the training data.

On Selecting a Single Frame Element By selecting only a single approximate critical point from the full frame $F_E(x)$ of a non-convex energy, canonicalization fundamentally becomes non-equivariant, and this cannot be overcome without extra assumptions. However, the overall output can still be approximately equivariant as long as canonicalization selects an element “close enough” to the training distribution: On the training distribution, f_{θ} must necessarily be approximately equivariant, as otherwise the training data are incompatible with equivariance to begin with. Then

$$g_x f_{\theta}(g_x^{-1}x) \approx g_x g_x^{-1} f_{\theta}(x) = f_{\theta}(x), \quad (13)$$

and approximate equivariance of the output is restored.

In practice, these effects remain sufficiently small to be negligible (see Section 4), allowing the overall network $\tilde{f}_{\theta}$ to maintain a high degree of approximate equivariance.

4. Experimental Results

We consider two different tasks: segmentation, where we desire diffeomorphism *equivariance*, and classification, where we aim for diffeomorphism *invariance*. We adopt data augmentation as our gold-standard baseline. While computationally demanding due to the infinite-dimensional nature of the diffeomorphism group, it allows sufficiently expressive models to learn the symmetry by densely sampling the group orbit. We do not aim to outperform this baseline’s accuracy, as it represents a theoretical performance ceiling for a given dataset; rather, we aim to match its performance without retraining the task model.

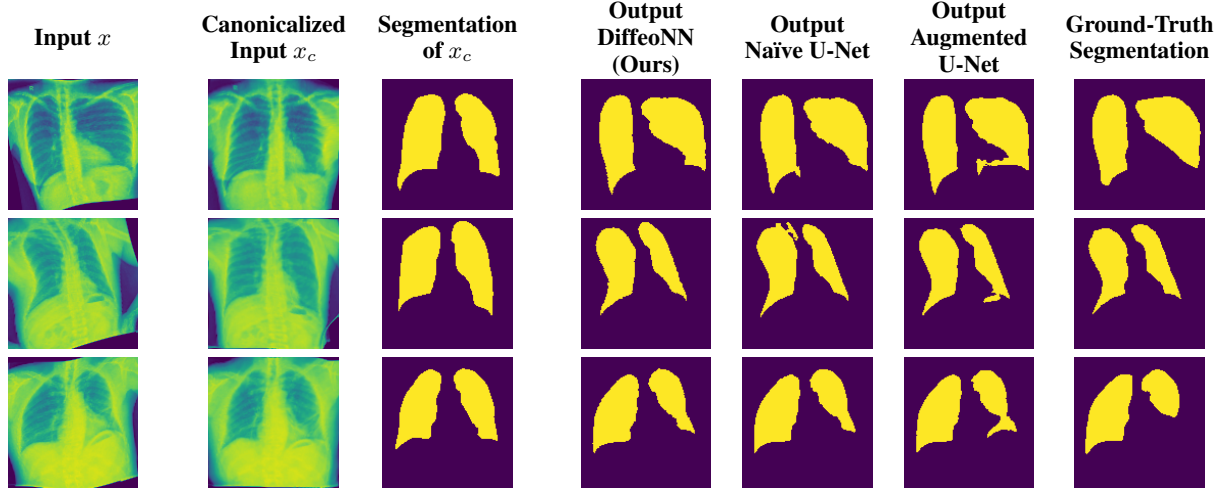


Figure 4. Lung segmentation of diffeomorphically transformed chest X-ray images from (Karmakar, 2024). Shown are the steps of DiffeoNN (columns two and three), as well as the final outputs of DiffeoNN, an augmented U-Net, and the inner U-Net of DiffeoNN without augmentation (naïve) (columns four, five, and six). The lung segmentations that are produced by DiffeoNN are closer to the ground truth than the segmentation outputs of the naïve and the augmented approach.

4.1. Segmentation

This work focuses on binary segmentation, i.e., the output is a binary mask $y_x : \Omega \rightarrow \{0, 1\}$. For segmentation, diffeomorphism equivariance means that if the input image is deformed by a diffeomorphism g , the predicted segmentation output should deform exactly the same way. Hence, the reverse canonicalization in (7) is exactly the inverse of the canonicalizing element g_x , i.e., $\tilde{f}_\theta(x) = g_x^{-1} \cdot \tilde{f}_\theta(g_x \cdot x)$.

4.1.1. SYNTHETIC DATASET

To visualize the functionality of DiffeoNN, we consider a synthetic toy example dataset of nested squares with the task of segmenting the inner square (Appendix A). The training dataset X_E contains images with two nested squares of varying sizes and added Gaussian noise, but crucially, restricted to squares aligned to the coordinate axes. For evaluation, we create a diffeomorphically transformed dataset X_{TE} by sampling deformation parameters from a radial basis function parameterization (Forti & Rozza, 2014) to warp the “canonical” images in X_E . Examples from the datasets X_E and X_{TE} can be found in Figure 5 and Appendix A.1.

Benchmarking Results To evaluate the performance of the proposed method, we compare the results of DiffeoNN, the simple inner U-Net of DiffeoNN (naïve approach), and an augmented U-Net that is trained in the same way as the inner U-Net of DiffeoNN but on an augmented dataset that includes “canonical” and transformed images. For more details of the setup and training, see Appendix A.2.

We apply the methods to 100 images from the test dataset X_{TE} and compute the Dice coefficient, Intersection-over-

Union (IoU), and pixel-wise accuracy (Table 1). DiffeoNN outperforms the naïve approach consistently. While DiffeoNN does not, on average, surpass the performance of our gold-standard baseline (augmented), it produces more stable and reliable results, with DiffeoNN exhibiting fewer outliers, suggesting that our approach is more robust than the augmented U-Net. Moreover, DiffeoNN achieves these results with a significantly smaller training dataset, making it a practical and efficient alternative to the data-intensive augmented U-Net.

4.1.2. LUNG SEGMENTATION

Additionally, we evaluate the performance of DiffeoNN on real-world data, using a dataset with chest X-ray images and their ground-truth lung segmentation from (Karmakar, 2024). The original dataset contains images and corresponding ground-truth segmentations, which are split into a training dataset of 504 image/segmentation pairs, a validation dataset of 100 pairs, and a test dataset of 100 pairs. We then proceed as in Section 4.1.1 to create a dataset of diffeomorphically transformed images.

Invariance of the Canonicalization A desirable property of the canonicalization step is diffeomorphism invariance. As the energy E_{can} is not strictly convex, the canonicalizing element g_x is not necessarily unique. Intuitively, the non-uniqueness is not problematic as long as $E_{\text{can}}(g_x \cdot x)$ is sufficiently small to indicate that the canonicalized input $x_c = g_x \cdot x$ is close enough to the training data X_E . To verify the approximate invariance in practice, we apply canonicalization to images from the training set X_E and their randomly transformed counterparts from X_{TE} . Ideally,

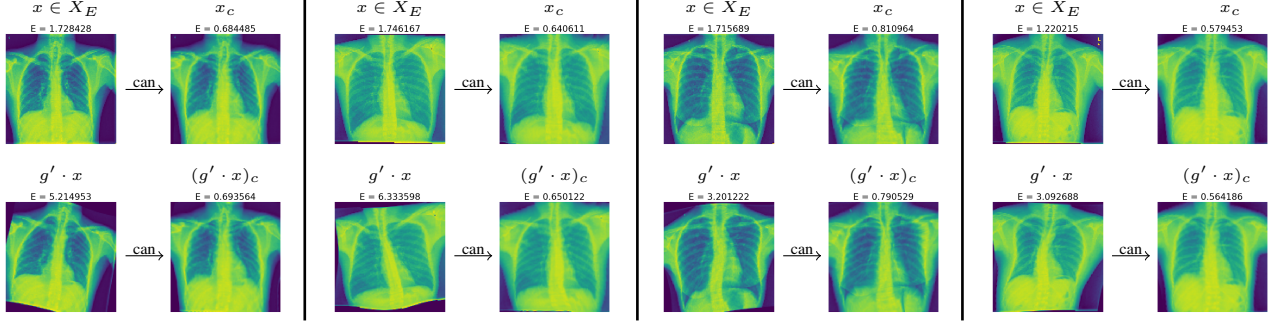


Figure 5. Examples of images $x \in X_E$ and their canonicalized form x_c (row one) with corresponding transformed images $g' \cdot x \in X_{TE}$ and their canonicalized form $(g' \cdot x)_c$ (row two) with their canonicalization energies. The experiment is described in Section 4.1.2. The energies before the canonicalization steps are very different. In contrast, after canonicalizing the inputs, the energies are approximately the same. Furthermore, the canonicalized images look very similar to images from X_E . This indicates the effectiveness of the canonicalization step and its invariance empirically.

the former should be kept almost invariant, and both canonicalizations should be close, confirmed by the experimental results (Figure 5). This is also reflected in the fact that the energies of the canonicalized elements are very similar.

Equivariance Error To quantify the empirical equivariance error of our practical framework, we randomly choose 100 images from the training dataset X_E and compare the segmentation output of the inner network f_θ , trained on X_E only, and the output of our network $\tilde{f}_\theta(x) = g_x^{-1} \cdot f_\theta(g_x \cdot x)$. If $\tilde{f}_\theta$ makes the inner network f_θ diffeomorphism-equivariant, the segmentation outputs should be identical. In our experiments, we obtain a mean Dice coefficient $\text{Dice}(f_\theta(x), \tilde{f}_\theta(x)) = 0.9812$ and a mean pixel-wise accuracy $\text{Acc}(f_\theta(x), \tilde{f}_\theta(x)) = 0.9902$. This indicates that the segmentations are in fact very similar and that the equivariance error remains sufficiently small in practice.

Benchmarking Results We apply DiffeoNN, the inner U-Net (naïve approach) and an augmented U-Net, which is trained on the original and the diffeomorphically transformed training dataset, to the diffeomorphically transformed images from the test dataset. Visual results are presented in Figure 4 and further details on the experimental setup, training, and additional examples in Appendix B.2. The results closely mirror those obtained on the synthetic dataset, see Table 1b. On average, DiffeoNN outperforms the naïve approach, validating the effectiveness of our approach. The canonicalization step pushes the input image towards the training dataset by moving the thorax to a more central position and aligning the shoulders, which makes it easier for the inner U-Net to predict an accurate lung segmentation. While our method performs slightly below the augmented gold-standard baseline, it offers more flexibility by not relying on data augmentation or retraining, making it a practical and robust solution in real-world applications.

We further show that the framework can be naturally extended to 3D *computed tomography* (CT) data (Wasserthal, 2023) using the pre-trained TotalSegmentator (Wasserthal et al., 2023), with preliminary results shown in Appendix G.

4.2. Classification

To illustrate the ability of our approach to achieve invariance, i.e., $\tilde{f}_\theta(x) = \tilde{f}_\theta(g \cdot x)$, we test it on the MNIST dataset (LeCun et al., 1998) for the task of determining the genus, i.e., counting topological holes. The typical classification task of identifying the written digit is not diffeomorphism invariant as, e.g., “6” can be mapped to “9” by a diffeomorphism, so they would be assigned to the same class. Similarly to Section 4.1, we transform the MNIST dataset X_E diffeomorphically to create a second dataset X_{TE} for testing.

Benchmarking Results We compare the performance of DiffeoNN to the naïve approach, i.e., the inner convolutional classifier for determining the genus and an augmented approach. The latter is trained, in analogy with other experiments on images, from the original and transformed MNIST. The mean accuracy over 100 transformed MNIST images can be found in Table 1c and visual examples are presented in Appendix F. Our canonicalization strategy noticeably improves the naïve approach, matching the performance of the augmented classifier, yet without retraining on the augmented data.

5. Conclusion

By combining Lie group theory with differentiable image registration, we have developed a robust energy-based canonicalization strategy for diffeomorphism equivariance. Our approach is highly data-efficient, achieving approximate equivariance using only a small dataset without data augmentation. The primary advantage of this approach

Table 1. Performance of **DiffoNN** compared to an augmentation baseline (**Aug.**) and the inner network of DiffoNN without augmentation (**Naïve**) for (a/d) **segmentation of the inner square** on 100 images from the **synthetic dataset** (Appendix A), (b/e) **lung segmentation** on 100 diffeomorphically transformed chest X-rays from (Karmakar, 2024), and (c/f) **homology classification** task on 100 diffeomorphically transformed MNIST (LeCun et al., 1998). DiffoNN outperforms the naïve approach on average and with less variable and thus more stable results. On MNIST (c), it matches the augmented approach’s average accuracy for the classification task. Although DiffoNN does not surpass the augmented baseline on average for segmentation, it produces smaller outliers or fewer outliers, suggesting that our approach is more robust than the augmented U-Net. Furthermore, DiffoNN has the advantage of needing to be trained only on the simple dataset X_E , reducing training time and computational resources while requiring no retraining.

(a) Mean Intersection-over-Union (IoU), Dice coefficient (Dice), and pixel-wise accuracy (Acc.) on the **synthetic dataset**.

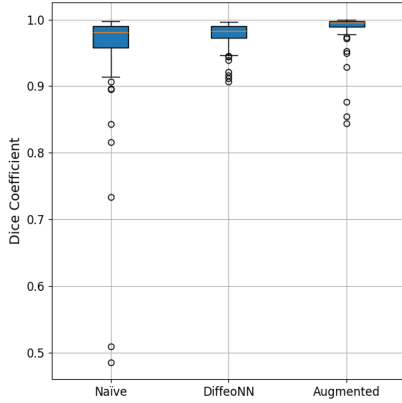
MODEL	IoU $\uparrow$	DICE $\uparrow$	ACC. $\uparrow$
NAÏVE	0.9276	0.9582	0.9962
DIFFOENN	0.9571	0.9777	0.9981
AUG.	0.9770	0.9878	0.9989

(b) Mean Intersection-over-Union (IoU), Dice coefficient (Dice), and pixel-wise accuracy (Acc.) on diffeomorphically transformed **chest X-rays**.

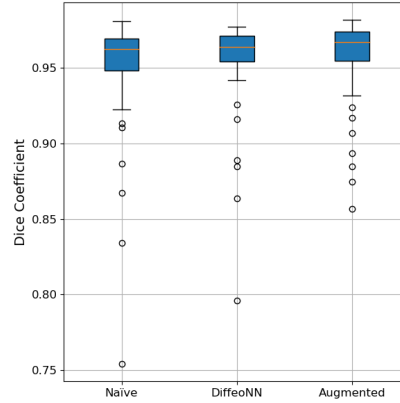
MODEL	IoU $\uparrow$	DICE $\uparrow$	ACC. $\uparrow$
NAÏVE	0.9147	0.9547	0.9755
DIFFOENN	0.9214	0.9586	0.9777
AUG.	0.9251	0.9606	0.9790

(c) Mean accuracy (Acc.) on the diffeomorphically transformed **MNIST** dataset.

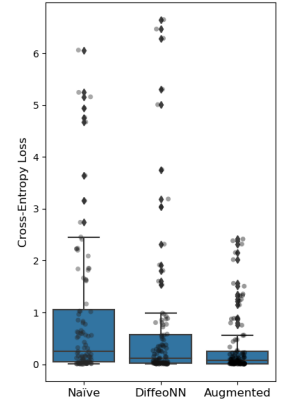
MODEL	ACC. $\uparrow$
NAÏVE	0.68
DIFFOENN	0.82
AUG.	0.82



(d) Box plots of the Dice coefficient of the **segmentation** results on the **synthetic dataset** X_{TE} .



(e) Box plots of the Dice coefficient on diffeomorphically transformed **chest X-rays**.



(f) Cross-entropy loss on transformed **MNIST**.

is its modularity: it can inject diffeomorphism equivariance into any pre-trained network without altering the inner task-performing model. Crucially, we demonstrated that energy canonicalization remains robust even when applied to groups as large as the group of diffeomorphisms.

Our results highlight a gap in the broader literature: despite accumulating empirical evidence of efficacy, there are few theoretical results regarding the stability and generalizability of canonicalization. Characterizing learning bounds and the sufficiency of approximate canonicalization for full equivariance remains a critical next step for future research.

Acknowledgments

We thank all collaborators and colleagues whose support and insights shaped this work, and we are grateful to the reviewers for their valuable feedback and constructive comments. We also thank the researchers who made their datasets and codebases publicly available. This work would not have been possible without these contributions.

ZS acknowledges support from the Cantab Capital Institute for the Mathematics of Information, Christs College and the Trinity Henry Barlow Scholarship scheme. CBS acknowledges support from the Royal Society Wolfson Fellowship, the EPSRC advanced career fellowship EP/V029428/1, the EPSRC programme grant EP/V026259/1, the Wellcome Innovator Awards 215733/Z/19/Z and 221633/Z/20/Z, the EPSRC funded ProBAI hub EP/Y028783/1, and the European Union Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement REMODEL.

Impact Statement

The work presented here is intended to advance the fields of Machine Learning and Geometric Deep Learning. We do not expect our work to have any specifically notable potential societal consequences apart from those usually associated with such advancements.

References

- Abramson, J., Adler, J., Dunger, J., Evans, R., Green, T., Pritzel, A., Ronneberger, O., Willmore, L., Ballard, A., Bambrick, J., Bodenstein, S., Evans, D., Hung, C.-C., O'Neill, M., Reiman, D., Tunyasuvunakool, K., Wu, Z., Žemgulytė, A., Arvaniti, E., and Jumper, J. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630:493–500, 05 2024. doi: 10.1038/s41586-024-07487-w.
- Arguillere, S., Trélat, E., Trouvé, A., and Younes, L. Shape deformation analysis from the optimal control viewpoint, 2014. URL <https://arxiv.org/abs/1401.0661>.
- Arjovsky, M., Chintala, S., and Bottou, L. Wasserstein generative adversarial networks. In Precup, D. and Teh, Y. W. (eds.), *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pp. 214–223. PMLR, 06–11 Aug 2017. URL <https://proceedings.mlr.press/v70/arjovsky17a.html>.
- Arsigny, V., Commowick, O., Pennec, X., and Ayache, N. A log-euclidean framework for statistics on diffeomorphisms. volume 9, pp. 924–31, 02 2006a. ISBN 978-3-540-44707-8. doi: 10.1007/11866565_113.
- Arsigny, V., Commowick, O., Pennec, X., and Ayache, N. A log-euclidean framework for statistics on diffeomorphisms. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pp. 924–931. Springer, 2006b.
- Ashburner, J. A fast diffeomorphic image registration algorithm. *NeuroImage*, 38(1):95–113, 2007. ISSN 1053-8119. doi: <https://doi.org/10.1016/j.neuroimage.2007.07.007>. URL <https://www.sciencedirect.com/science/article/pii/S1053811907005848>.
- Balakrishnan, G., Zhao, A., Sabuncu, M. R., Guttag, J., and Dalca, A. V. VoxelMorph: a learning framework for deformable medical image registration. *IEEE Transactions on Medical Imaging*, 38(8):1788–1800, 2019. doi: 10.1109/TMI.2019.2897538.
- Bekkers, E. J. B-spline CNNs on Lie groups, 2021. URL <https://arxiv.org/abs/1909.12057>.
- Bekkers, E. J., Lafarge, M. W., Veta, M., Eppenhof, K. A., Pluim, J. P., and Duits, R. Roto-translation covariant convolutional networks for medical image analysis, 2018. URL <https://arxiv.org/abs/1804.03393>.
- Benning, M. and Burger, M. Modern Regularization Methods for Inverse Problems, January 2018. URL <http://arxiv.org/abs/1801.09922>. arXiv:1801.09922 [math].
- Bietti, A., Venturi, L., and Bruna, J. On the sample complexity of learning under invariance and geometric stability, 2021. URL <https://arxiv.org/abs/2106.07148>.
- Bostelmann, J., Gildemeister, O., and Lellmann, J. Stationary velocity fields on matrix groups for deformable image registration. *Journal of Mathematical Imaging and Vision*, 68:9, 2026. ISSN 1573-7683. doi: 10.1007/s10851-026-01284-y. URL <https://doi.org/10.1007/s10851-026-01284-y>.
- Brehmer, J., Behrends, S., de Haan, P., and Cohen, T. Does equivariance matter at scale?, 2025. URL <https://arxiv.org/abs/2410.23179>.
- Bronstein, M. M., Bruna, J., Cohen, T., and Velicković, P. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges, 2021. URL <https://arxiv.org/abs/2104.13478>.
- Chen, J., Frey, E. C., He, Y., Segars, W. P., Li, Y., and Du, Y. Transmorph: Transformer for unsupervised medical image registration. *Medical image analysis*, 82:102615, 2022a.
- Chen, X., Wang, X., Zhang, K., Fung, K.-M., Thai, T. C., Moore, K., Mannel, R. S., Liu, H., Zheng, B., and Qiu, Y. Recent advances and clinical applications of deep learning in medical image analysis. *Medical Image Analysis*, 79, 2022b. ISSN 1361-8415. doi: 10.1016/j.media.2022.102444. URL <http://dx.doi.org/10.1016/j.media.2022.102444>.
- Cohen, T., Geiger, M., Koehler, J., and Welling, M. Spherical CNNs. 01 2018. doi: 10.48550/arXiv.1801.10130.
- Cohen, T. S. and Welling, M. Group equivariant convolutional networks, 2016. URL <https://arxiv.org/abs/1602.07576>.
- Dalca, A. V., Balakrishnan, G., Guttag, J., and Sabuncu, M. R. Unsupervised learning of probabilistic diffeomorphic registration for images and surfaces. *Medical image analysis*, 57:226–236, 2019.
- Degrave, J., Felici, F., Buchli, J., Neunert, M., Tracey, B., Carpanese, F., Ewalds, T., Hafner, R., Abdolmaleki, A., Casas, D., Donner, C., Fritz, L., Galperti, C., Huber, A., Keeling, J., Tsimpoukelli, M., Kay, J., Merle, A., Moret, J.-M., and Riedmiller, M. Magnetic control of tokamak plasmas through deep reinforcement learning. *Nature*, 602:414–419, 02 2022. doi: 10.1038/s41586-021-04301-9.
- Dieleman, S., Willett, K. W., and Dambre, J. Rotation-invariant convolutional neural networks for galaxy morphology prediction. *Monthly Notices of the Royal Astronomical Society*, 450(2):1441–1459, April 2015. ISSN

- 0035-8711. doi: 10.1093/mnras/stv632. URL <http://dx.doi.org/10.1093/mnras/stv632>.
- Dym, N., Lawrence, H., and Siegel, J. W. Equivariant frames and the impossibility of continuous canonicalization, 2024. URL <https://arxiv.org/abs/2402.16077>.
- Eijkelboom, F., Bekkers, E. J., Sonke, J.-J., and Worring, M. E(n) equivariant topological neural networks. *arXiv preprint arXiv:2405.15429*, 2024.
- Ferrari, L., Frosini, P., Quercioli, N., and Tombari, F. A topological model for partial equivariance in deep learning and data analysis. *Frontiers in Artificial Intelligence*, 6:1303931, 2023.
- Finzi, M., Stanton, S., Izmailov, P., and Wilson, A. G. Generalizing convolutional neural networks for equivariance to Lie groups on arbitrary continuous data, 2020. URL <https://arxiv.org/abs/2002.12880>.
- Forti, D. and Rozza, G. Efficient geometrical parametrisation techniques of interfaces for reduced-order modelling: application to fluid–structure interaction coupling problems1. *International Journal of Computational Fluid Dynamics*, 28(3-4):158–169, 2014. doi: 10.1080/10618562.2014.932352. URL <https://doi.org/10.1080/10618562.2014.932352>.
- Fukushima, K. and Miyake, S. Neocognitron: A new algorithm for pattern recognition tolerant of deformations and shifts in position. *Pattern Recognit.*, 15:455–469, 1982. URL <https://api.semanticscholar.org/CorpusID:2357880>.
- Goodfellow, I., Bengio, Y., and Courville, A. *Deep Learning*. MIT Press, 2016. URL <http://www.deeplearningbook.org>. Book in preparation for MIT Press.
- Hajij, M., Zamzmi, G., Papamarkou, T., Miolane, N., Guzmán-Sáenz, A., Ramamurthy, K. N., Birdal, T., Dey, T. K., Mukherjee, S., Samaga, S. N., et al. Topological deep learning: Going beyond graph data. *arXiv preprint arXiv:2206.00606*, 2023.
- Han, K., Sun, S., Yan, X., You, C., Tang, H., Naushad, J., Ma, H., Kong, D., and Xie, X. Diffeomorphic image registration with neural velocity field. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 1869–1879, 2023.
- Hauberg, S., Freifeld, O., Larsen, A. B. L., III, J. W. F., and Hansen, L. K. Dreaming more data: Class-dependent distributions over diffeomorphisms for learned data augmentation, 2016. URL <https://arxiv.org/abs/1510.02795>.
- He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016. doi: 10.1109/CVPR.2016.90.
- Hernandez, M., Olmos Gassó, S., and Pennec, X. Comparing algorithms for diffeomorphic registration: Stationary lddmm and diffeomorphic demons. In Pennec, X. and Joshi, S. (eds.), *2nd MICCAI Workshop on Mathematical Foundations of Computational Anatomy (MFCA)*, pp. 24–35, New York, NY, USA, 2008.
- Kaba, S.-O., Mondal, A. K., Zhang, Y., Bengio, Y., and Ravanbakhsh, S. Equivariance with learned canonicalization functions, 2023. URL <https://arxiv.org/abs/2211.06489>.
- Karmakar, T. Chest X-ray dataset for lung segmentation. Kaggle Dataset, 2024. URL <https://www.kaggle.com/datasets/iamtapendu/chest-x-ray-lungs-segmentation>. Accessed: 13 February 2026.
- Kingma, D. P. and Ba, J. Adam: A method for stochastic optimization, 2017. URL <https://arxiv.org/abs/1412.6980>.
- Kingma, D. P. and Welling, M. Auto-encoding variational bayes, 2022. URL <https://arxiv.org/abs/1312.6114>.
- Kondor, R. and Trivedi, S. On the generalization of equivariance and convolution in neural networks to the action of compact groups, 2018. URL <https://arxiv.org/abs/1802.03690>.
- Krizhevsky, A., Sutskever, I., and Hinton, G. E. Imagenet classification with deep convolutional neural networks. In Pereira, F., Burges, C., Bottou, L., and Weinberger, K. (eds.), *Advances in Neural Information Processing Systems*, volume 25. Curran Associates, Inc., 2012. URL https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf.
- Kuang, D. On reducing negative jacobian determinant of the deformation predicted by deep registration networks, 2019. URL <https://arxiv.org/abs/1907.00068>.
- Kumar, H., Parada-Mayorga, A., and Ribeiro, A. Lie group algebra convolutional filters, 2024. URL <https://arxiv.org/abs/2305.04431>.
- Lawrence, H., Hofgard, E., Chen, Y., Smidt, T., and Walters, R. Detecting symmetry-breaking in molecular

- data distributions. In *AI for Accelerated Materials Design Workshop at ICLR 2025*, 2025a. URL <https://iclr.cc/virtual/2025/32950>. Poster paper.
- Lawrence, H., Portilheiro, V., Zhang, Y., and Kaba, S.-O. Improving equivariant networks with probabilistic symmetry breaking, 2025b. URL <https://arxiv.org/abs/2503.21985>.
- LeCun, Y., Boser, B., Denker, J. S., Henderson, D., Howard, R. E., Hubbard, W., and Jackel, L. D. Backpropagation applied to handwritten zip code recognition. *Neural Computation*, 1(4):541–551, 1989. doi: 10.1162/neco.1989.1.4.541.
- LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. Gradient-based learning applied to document recognition. *Proc. IEEE*, 86:2278–2324, 1998. URL <https://api.semanticscholar.org/CorpusID:14542261>.
- Li, S., Song, W., Fang, L., Chen, Y., Ghamisi, P., and Benediktsson, J. A. Deep learning for hyperspectral image classification: An overview. *IEEE Transactions on Geoscience and Remote Sensing*, 57(9):6690–6709, 2019. doi: 10.1109/TGRS.2019.2907932.
- Liu, C., Liao, D., Parada-Mayorga, A., Ribeiro, A., DiStasio, M., and Krishnaswamy, S. Diffkillr: Killing and recreating diffeomorphisms for cell annotation in dense microscopy images. In *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5. IEEE, 2025.
- Liu, Y., Chen, J., Wei, S., Carass, A., and Prince, J. On finite difference jacobian computation in deformable image registration, 2023. URL <https://arxiv.org/abs/2212.06060>.
- Lunz, S., Öktem, O., and Schönlieb, C.-B. Adversarial regularizers in inverse problems, 2019. URL <https://arxiv.org/abs/1805.11572>.
- Ma, G., Wang, Y., Lim, D., Jegelka, S., and Wang, Y. A Canonicalization Perspective on Invariant and Equivariant Learning, November 2024. URL <http://arxiv.org/abs/2405.18378>. arXiv:2405.18378.
- Marcus, G. Deep learning: A critical appraisal, 2018. URL <https://arxiv.org/abs/1801.00631>.
- Miller, M. Computational anatomy: Shape, growth, and atrophy comparison via diffeomorphisms. *NeuroImage*, 23 Suppl 1:S19–33, 02 2004. doi: 10.1016/j.neuroimage.2004.07.021.
- Milnor, J. W. Remarks on infinite-dimensional Lie groups. In DeWitt, B. S. and Stora, R. (eds.), *Relativity, Groups and Topology II*. North-Holland, 1984.
- Moskowitz, M. and Sacksteder, R. The exponential map and differential equations on real Lie groups. *Journal of Lie Theory*, 13(1):291–306, 2003.
- Mukherjee, S., Dittmer, S., Shumaylov, Z., Lunz, S., Öktem, O., and Schönlieb, C.-B. Data-driven convex regularizers for inverse problems. pp. 13386–13390, 04 2024. doi: 10.1109/ICASSP48485.2024.10447719.
- Murphy, R. L., Srinivasan, B., Rao, V., and Ribeiro, B. Relational pooling for graph representations, 2019. URL <https://arxiv.org/abs/1903.02541>.
- Omori, H. *Infinite Dimensional Lie Transformation Groups*, volume 427 of *Lecture Notes in Mathematics*. Springer, Berlin, Heidelberg, 1974. ISBN 978-3-540-07013-9.
- Onieva Onieva, J., Marti-Fuster, B., Pedrero de la Puente, M., and San José Estépar, R. Diffeomorphic lung registration using deep CNNs and reinforced learning. In Stoyanov, D., Taylor, Z., Kainz, B., Maicas, G., Beichel, R. R., Martel, A., Maier-Hein, L., Bhatia, K., Vercauteren, T., Oktay, O., Carneiro, G., Bradley, A. P., Nascimento, J., Min, H., Brown, M. S., Jacobs, C., Lassen-Schmidt, B., Mori, K., Petersen, J., San José Estépar, R., Schmidt-Richberg, A., and Veiga, C. (eds.), *Image Analysis for Moving Organ, Breast, and Thoracic Images*, volume 11040 of *Lecture Notes in Computer Science*, pp. 284–294, Cham, 2018. Springer International Publishing. doi: 10.1007/978-3-030-00946-5_28.
- Puny, O., Atzmon, M., Ben-Hamu, H., Misra, I., Grover, A., Smith, E. J., and Lipman, Y. Frame averaging for invariant and equivariant network design, 2022. URL <https://arxiv.org/abs/2110.03336>.
- Radiologi RSUA. Rsua chest X-ray dataset, 2023. URL <https://doi.org/10.17632/2jg8vfdmpm.1>.
- Ronneberger, O., Fischer, P., and Brox, T. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pp. 234–241. Springer, 2015.
- Shorten, C. and Khoshgoftaar, T. A survey on image data augmentation for deep learning. *Journal of Big Data*, 6, 07 2019. doi: 10.1186/s40537-019-0197-0.
- Shumaylov, Z., Zaika, P., Rowbottom, J., Sherry, F., Weber, M., and Schönlieb, C.-B. Lie algebra canonicalization: Equivariant neural operators under arbitrary Lie groups, 10 2024.
- Sitzmann, V., Martel, J., Bergman, A., Lindell, D., and Wetzstein, G. Implicit neural representations with periodic activation functions. *Advances in neural information processing systems*, 33:7462–7473, 2020.

- Sotiras, A., Davatzikos, C., and Paragios, N. Deformable medical image registration: A survey. *IEEE Transactions on Medical Imaging*, 32(7):1153–1190, 2013. doi: 10.1109/TMI.2013.2265603.
- Sun, C., Shrivastava, A., Singh, S., and Gupta, A. Revisiting unreasonable effectiveness of data in deep learning era, 2017. URL <https://arxiv.org/abs/1707.02968>.
- Tahmasebi, B. and Jegelka, S. Generalization bounds for canonicalization: A comparative study with group averaging. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=n0lXaskyk5>.
- Trouvé, A. Diffeomorphisms groups and pattern matching in image analysis. *International journal of computer vision*, 28(3):213–221, 1998.
- Voulodimos, A., Doulamis, N., Doulamis, A., and Protopadakis, E. Deep learning for computer vision: A brief review. *Computational Intelligence and Neuroscience*, 2018:1–13, 02 2018. doi: 10.1155/2018/7068349.
- Wang, T., Lei, Y., Fu, Y., Curran, W. J., Liu, T., and Yang, X. Medical imaging synthesis using deep learning and its clinical applications: A review, 2020. URL <https://arxiv.org/abs/2004.10322>.
- Wasserthal, J. Dataset with segmentations of 117 important anatomical structures in 1228 CT images, 2023. URL <https://doi.org/10.5281/zenodo.10047292>.
- Wasserthal, J., Breit, H.-C., Meyer, M. T., Pradella, M., Hinck, D., Sauter, A. W., Heye, T. J., Boll, D. A., Cyriac, J., Shan, Y., Bach, M., and Segeroth, M. Totalsegmentator: Robust segmentation of 104 anatomic structures in CT images. *Radiology: Artificial Intelligence*, 5(5): e230024, 2023. doi: 10.1148/ryai.230024.
- Weijler, L. and Hermosilla, P. Efficient continuous group convolutions for local SE(3) equivariance in 3D point clouds, 2025. URL <https://arxiv.org/abs/2502.07505>.
- Wirgin, A. The inverse crime, 2004. URL <https://arxiv.org/abs/math-ph/0401050>.
- Younes, L. *Shapes and Diffeomorphisms*, volume 171 of *Applied Mathematical Sciences*. Springer, Berlin, Heidelberg, 2010. ISBN 978-3-642-12054-1. doi: 10.1007/978-3-642-12055-8.
- Zhang, C., Bengio, S., Hardt, M., Recht, B., and Vinyals, O. Understanding deep learning requires rethinking generalization, 2017. URL <https://arxiv.org/abs/1611.03530>.
- Zhao, Z.-Q., Zheng, P., Xu, S.-t., and Wu, X. Object detection with deep learning: A review. *IEEE transactions on neural networks and learning systems*, 30(11):3212–3232, 2019.

A. Synthetic Dataset

A.1. Examples of Synthetic Dataset

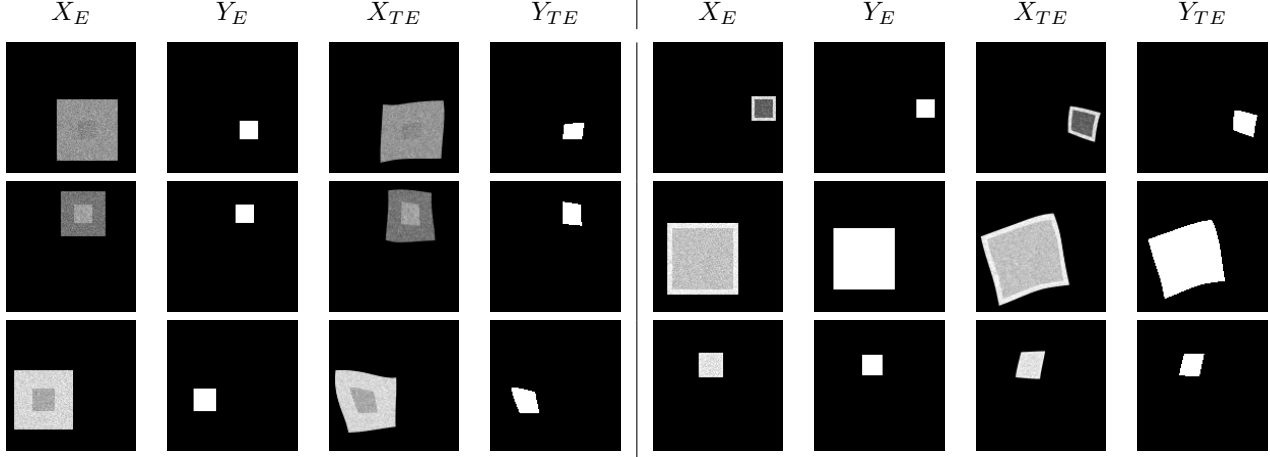


Figure 6. Examples of the generated square images X_E (column one) and their binary segmentation Y_E (column two), as well as a random diffeomorphic transformations of those images X_{TE} , (column three) and the corresponding output Y_{TE} (column four). The images in X_E contain two nested squares of varying size and color, where the squares’ edges are parallel to the image edges with added Gaussian noise. The binary segmentation Y_E indicates the inner square. The images X_{TE} and their corresponding segmentations Y_{TE} are obtained by transforming the images X_E and their binary segmentations Y_E with the same randomly chosen transformations.

For our experiments, we randomly generate images of nested squares X_E and corresponding binary segmentation of the inner square Y_E . To evaluate the performance of our method, the invariance of the canonicalization step and equivariance of the whole framework, we generate diffeomorphically transformed data by randomly transforming image and segmentation pairs from X_E and Y_E . To avoid an “inverse crime” (Wirgin, 2004), we do not use the same technique as in our canonicalization step to parameterize diffeomorphisms for the creation of our diffeomorphically transformed synthetic data. The implementation of these transformations relies on the implementation of (Bostelmann et al., 2026).

The transformations are parameterized using thin-plate spline radial basis function interpolation based on a set of randomly sampled control points. In all experiments, 12 control points are used and displaced with a maximum magnitude of 0.12 in normalized image coordinates. In addition, random rotations are incorporated to increase variability and generate more realistic non-rigid deformations. The resulting deformation field is converted into a dense vector field and applied jointly to the image and segmentation mask using spatial resampling. Bilinear interpolation is used for image intensities, while nearest-neighbor interpolation is employed for segmentation masks in order to preserve binary labels.

To reduce unrealistic artifacts, the determinant of the Jacobian of the deformation field is monitored to identify non-diffeomorphic transformations. Moreover, samples in which foreground structures intersect the image boundary are rejected.

A.2. Experiments on the Synthetic Dataset

For the inner network, we train a simple U-Net (Ronneberger et al., 2015) with a learning rate of $5 \cdot 10^{-5}$ and a batch size of 2 for 20 epochs on the training dataset X_E using Adam (Kingma & Ba, 2017).

For the image similarity energy (see Section 3.2 and Appendix D) a VAE-CNN with a ten-dimensional latent space is trained on X_E using Adam (Kingma & Ba, 2017) with a batch size of 1, a learning rate of $5 \cdot 10^{-5}$ and 50 epochs, following the implementation of (Kingma & Welling, 2022; Shumaylov et al., 2024). The implementation for the convex adversarial discriminator closely follows (Lunz et al., 2019; Mukherjee et al., 2024; Shumaylov et al., 2024). The images are transformed on-the-fly with randomly chosen diffeomorphisms similar to the ones, used for creating X_{TE} . The gradient penalty weight is set to $\mu = 10$ and the learning rate to $1 \cdot 10^{-4}$, the batch size to 16, and the number of epochs to 50. The optimizer is Adam (Kingma & Ba, 2017). See also <https://github.com/mic-uzl/DiffeoNN>.

We set the scalar weights in E_{can} , see (10) and (11), to $\lambda_{\text{adv}} = 0.01$, $\lambda_{\text{VAE}} = 1 \cdot 10^{-5}$, $\lambda_{\Delta} = 1$, and $\lambda_{\mathcal{J}} = 10$.

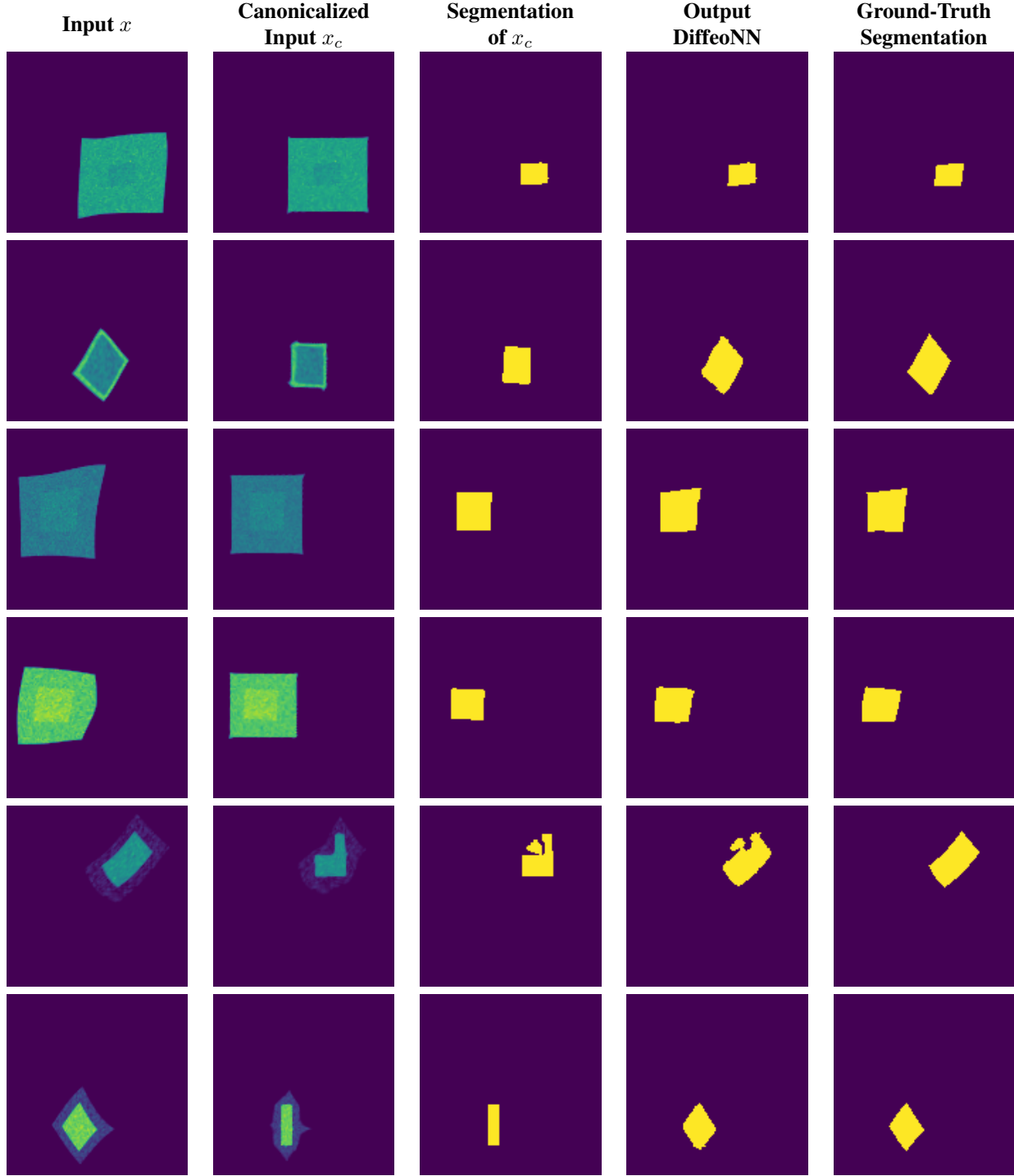


Figure 7. Examples of results of DiffeoNN step by step on the synthetic test dataset X_{TE} . The **first four rows** show a nearly perfect segmentation through DiffeoNN. The canonicalized input images look very similar to the training dataset of squares X_E , and the output segmentation is very similar to the ground-truth segmentation. **Row five** shows an example where the canonicalization step failed. As a result, the segmentation of the canonicalized image x_c contains artifacts. The artifacts are transferred to the output segmentation of the input x , which differs noticeably from the ground-truth segmentation. In **row six**, we show an example where the canonicalization step leads to an image x_c that contains rectangles rather than squares. Even though this image does not look exactly like the training images with squares X_E , the segmentation output for x_c is still very accurate, which leads to a segmentation output for x that is nearly identical to the ground truth.



Figure 8. Examples comparing the results of the naïve U-Net, DiffoNN, and the augmented U-Net on X_{TE} (synthetic dataset). **Column one** shows the input image from $x \in X_{TE}$, and **column five** shows the corresponding ground-truth segmentation. In **columns two to four**, we present the segmentation output of x by different networks. In the **first four rows**, the segmentation of the naïve U-Net is noticeably different from the ground truth, while the segmentations of DiffoNN and the augmented U-Net are similar. The **last two rows** show examples where the augmented U-Net performs worse than the other two methods.

B. 2D Lung Segmentation

B.1. Implementation Details

Our training dataset consists of 504 chest X-ray image/lung segmentation pairs from (Karmakar, 2024), while the validation dataset contains 100 pairs. The images are resized to 128×128 grayscale images with corresponding binary segmentation masks of the same resolution.

Transformed Images for Testing The diffeomorphically transformed image/segmentation pairs are created similarly to the transformed synthetic data, see Appendix A.1: Each image and mask is transformed using a deformation field generated through thin-plate spline radial basis function (RBF) interpolation. The deformation is defined by 12 control points uniformly placed across the image domain in normalized coordinates $[-0.5, 0.5]^2$. For each control point, a random displacement vector with maximum norm 0.12 is sampled. In addition, a random rotation with angle $\theta \sim \mathcal{U}(-\pi/6, \pi/6)$ is applied. No translation is used, making it less likely that transformed structures move outside the image frame. The resulting displacement vector field (DVF) is used to spatially warp both the image and its corresponding segmentation mask using bilinear interpolation. To preserve binary labels, transformed masks are rounded to $\{0, 1\}$.

Although the use of RBFs does not guarantee a diffeomorphic transformation, we make it “smooth enough” through a suitable choice of parameters, making extreme deformations highly unlikely. Furthermore, the Jacobian determinant of the DVF is evaluated to detect non-orientation-preserving deformations.

To ensure that foreground structures remain within the image domain, we verify that no non-background point touches the image boundary after the transformation. Otherwise, the transformation is discarded and a new one is randomly created.

U-Net Lung segmentation was performed using a 2D U-Net, implemented as a lightly modified version of the original U-Net architecture (Ronneberger et al., 2015). The *contracting path* consists of convolutional blocks followed by max pooling, while the *expanding path* uses transposed convolutions for upsampling. Each block contains two 3×3 convolutions followed by batch normalization and ReLU activations.

The network is trained using the Adam optimizer (Kingma & Ba, 2017) for 20 epochs with a learning rate of $5 \cdot 10^{-4}$ and batch size 2. Training is supervised using a pixel-wise cross-entropy loss, and performance is evaluated on the validation dataset.

The augmented U-Net is trained using the same hyperparameters on an augmented dataset consisting of the original 504 image/segmentation pairs along with diffeomorphically transformed versions. The transformations are sampled randomly in a manner similar to those, used to generate the transformed test dataset.

Canonicalization Energy For the image similarity energy (see Section 3.2 and Appendix D), a VAE-CNN with a 100-dimensional latent space is trained on the training dataset using Adam (Kingma & Ba, 2017), with batch size 1, learning rate $1 \cdot 10^{-4}$, and 50 training epochs, following the implementation of (Kingma & Welling, 2022; Shumaylov et al., 2024).

The implementation of the convex adversarial discriminator closely follows (Lunz et al., 2019; Mukherjee et al., 2024; Shumaylov et al., 2024). During training, images are transformed on the fly using randomly sampled diffeomorphisms, similar to those used to construct the transformed test dataset. The discriminator is trained with gradient penalty weight $\mu = 10$, learning rate $1 \cdot 10^{-4}$, batch size 16, and 100 epochs.

In the canonicalization step, we set weights in the canonicalization energy E_{can} to $\lambda_{\text{adv}} = 0.001$, $\lambda_{\text{VAE}} = 0.001$, $\lambda_{\Delta} = 1$, and $\lambda_{\mathcal{J}} = 10$ and perform 100 gradient steps to find a canonicalizing element. For the source code, see <https://github.com/mic-uzl/DiffeoNN>.

B.2. Visual Results

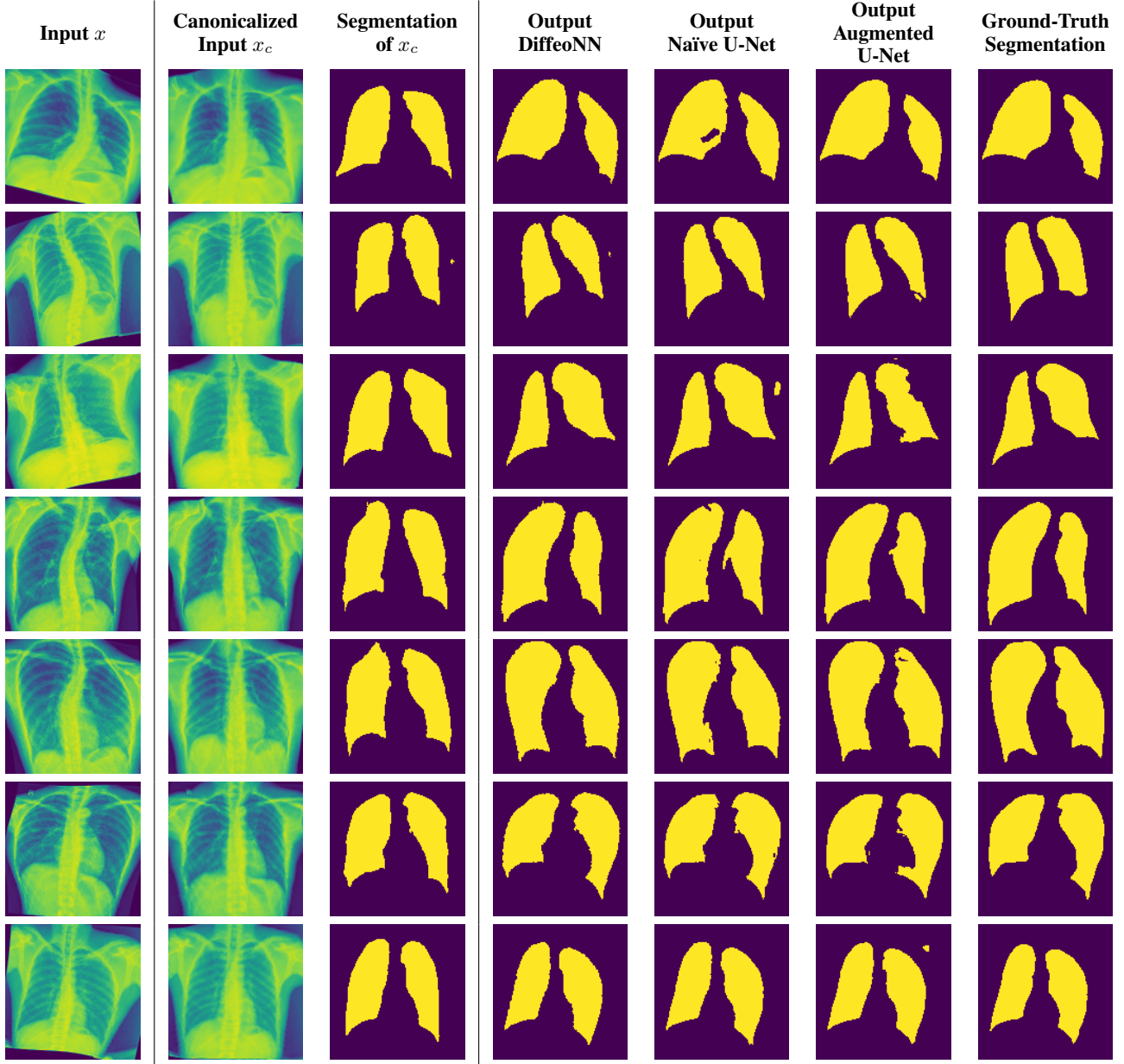


Figure 9. Lung segmentation of diffeomorphically transformed chest X-ray images from (Karmakar, 2024). Shown are the steps of DiffeoNN (**columns two and three**), as well as the final outputs of DiffeoNN, the inner U-Net of DiffeoNN without augmentation (naïve), and an augmented U-Net (**columns four, five, and six**). In the canonicalized images, the thorax is moved to a more central position and the shoulders were aligned, which makes it easier for the inner U-Net to predict an accurate lung segmentation. If the canonicalization step works, DiffeoNN produces more accurate lung segmentations than the naïve approach. The segmentation output of DiffeoNN is very similar to the output of the augmented U-Net (and the ground truth) and occasionally even provides a more accurate segmentation (see the last row).

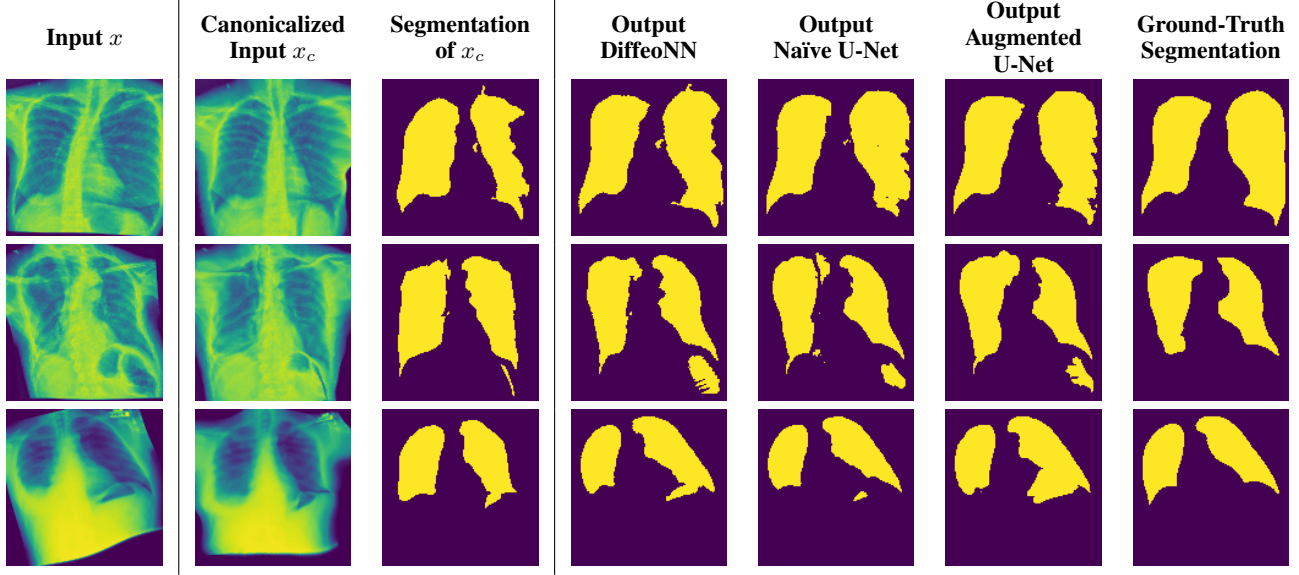


Figure 10. Lung segmentation of diffeomorphically transformed chest X-ray images from (Karmakar, 2024) on which neither of the three methods produced satisfactory lung segmentations. The canonicalized images do look “close” to the training dataset X_E , but there are artifacts and tearings in the lung area. The final segmentation output of DiffeoNN does not match the ground-truth segmentation. However, both the naïve and the augmented approach also struggle with these specific inputs.

B.3. Exaggerated Diffeomorphisms

To explore the robustness and generalizability of our method further, we evaluate DiffeoNN on images that differed more extremely from the training dataset. For the experiment, we create the transformed testing image/segmentation pairs as before (see Appendix B.1), but set the maximum norm of the randomly drawn displacement vector to 0.6. This results in more exaggerated transformations. We compare the performance of DiffeoNN to the performance of the inner U-Net (naïve) and the augmented U-Net. Exemplary visual results can be found in Figure 12 and the Dice coefficients in Figure 11. Although the canonicalization step does not always result in perfect canonicalized images that are “close” to the training samples, DiffeoNN outperforms not only the naïve, but also the augmented approach.

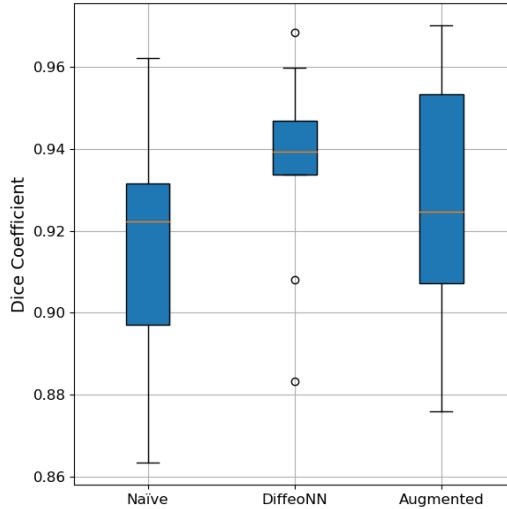


Figure 11. Dice coefficient on transformed chest X-rays with exaggerated diffeomorphisms. Overall, DiffeoNN produces more accurate lung segmentations than the naïve and the augmented U-Net, as the mean Dice Coefficient (orange line) is higher and DiffeoNN has fewer drastic outliers.

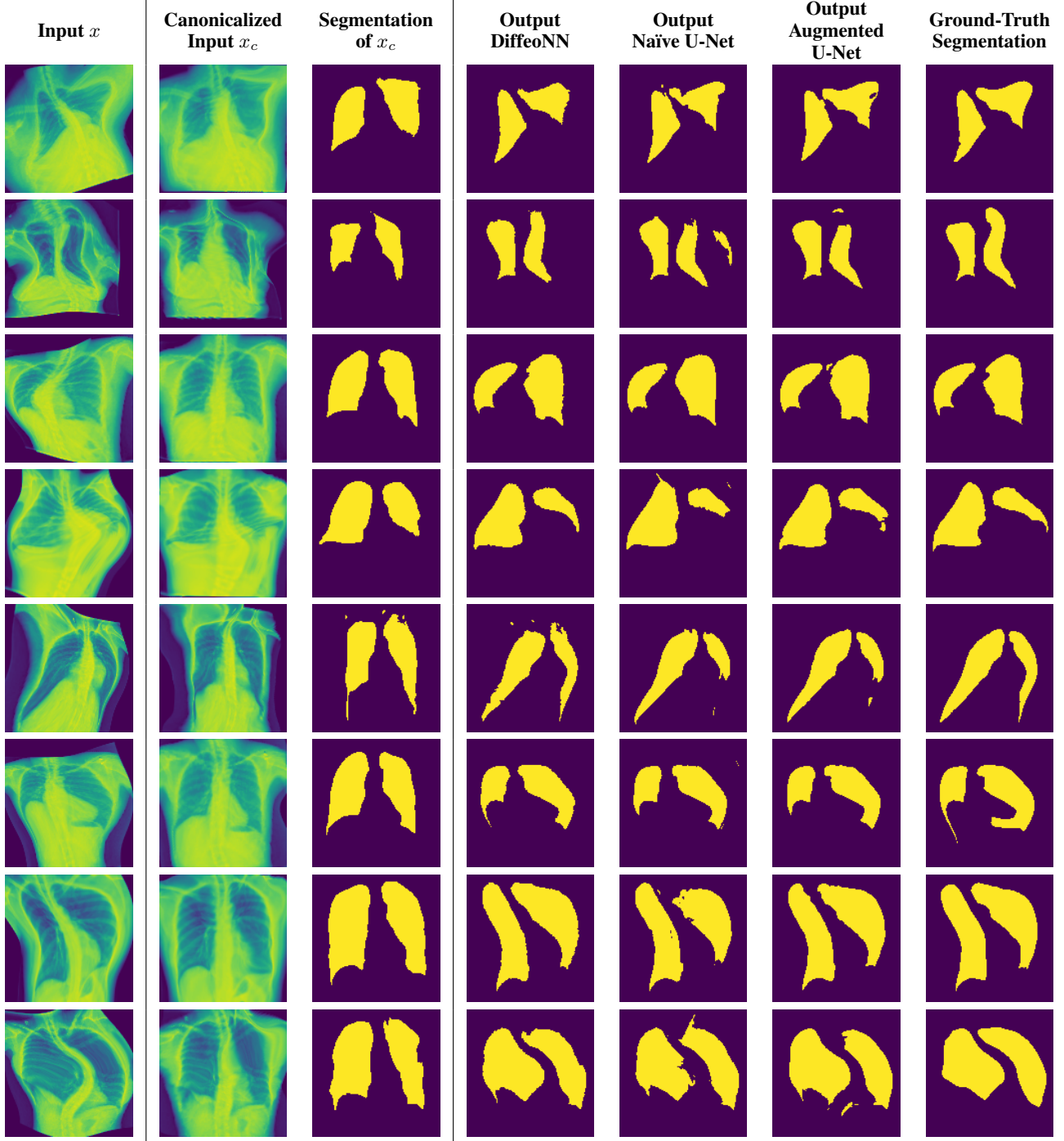


Figure 12. Lung segmentation of chest X-ray images from (Karmakar, 2024), where the data (chest X-ray and ground-truth segmentation) have undergone exaggerated transformations. Although the canonicalization step occasionally creates wavy warping artifacts, DiffeoNN produces satisfactory lung segmentations, which are more accurate than the naïve and the augmented approach and have fewer random artifacts than the other two approaches.

C. SVF-generated Diffeomorphisms

Unlike classical equivariance methods, the parameterization of the canonicalization group enters only in the optimization step. It easily tolerates coarse parameterizations, or those that lack closure under composition, provided that approximate minimizers can be found. Especially for learning-based image registration, a parameterization with stationary velocity fields is commonly used (Balakrishnan et al., 2019; Chen et al., 2022a; Han et al., 2023) due to the fast computation offered by the scaling-and-squaring approach (Arsigny et al., 2006b). As the group of diffeomorphisms is disconnected and separates into the two connected sets of orientation-preserving and orientation-reversing diffeomorphisms, and flow-based methods start from the identity, we can only expect to achieve orientation-preserving diffeomorphisms. This is not a specific issue with using SVFs; it also applies to time-dependent velocity fields used in LDDMMs (Younes, 2010) and may even be preferred. Orientation-preserving diffeomorphisms $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, which cannot be obtained by integration of stationary velocity fields in the continuous setting, include, for example:

$$\phi(x) := \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} x,$$

or more generally, linear deformations of $\mathbb{R}^2$ with two different negative eigenvalues. It is straightforward to create similar examples for the 3-dimensional case. In such cases, the matrix logarithm of the generating matrix includes imaginary components, which directly shows that no SVF of the form $v(x) := Mx$ can create the corresponding action of this group element ϕ by integrating the velocity field, as the (matrix-)exponential map is exactly the solution of integrating a stationary linear velocity field at time 1 starting from the identity. To the best of our knowledge, no continuous, possibly nonlinear, stationary velocity field can generate this deformation. If we wanted to include such deformations, it would be an easy adaptation to concatenate two diffeomorphisms obtained by integrating SVFs, thereby finding a compromise between the fast computability of SVFs and the expressivity of time-dependent velocity fields. For the example above, this allows factoring the matrix

$$\begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Both matrices on the right-hand side have real matrix logarithms, so that even linear SVFs can create these individual deformations. See also the theorem of Moskowitz–Sacksteder for a deeper theoretical and more general connection to the exponential map of Lie groups (Moskowitz & Sacksteder, 2003).

Ablation Study To verify the expressiveness of parameterizing the group of diffeomorphisms via SVFs, we compared the canonicalization step with a deformation parameterization via a displacement field defined by

$$\phi_u(x) := x + u(x), \tag{14}$$

while fixing the rest of our canonicalization step. In this way, only the difference from the identity id is generated from the model. By default, displacement fields automatically parameterize every transformation ϕ by setting $u = \phi - \text{id}$. However, these parameterizations do not inherently guarantee a diffeomorphism, and even if the resulting deformation is diffeomorphic, computing the inverse is nontrivial and requires solving an additional optimization problem. In contrast, SVFs provide the inverse transformation at almost no cost by simply integrating the negative velocity field. Visual results on the chest X-ray dataset (Karmakar, 2024) are presented in Figure 13. Our results show that the canonicalized images closely resemble each other regardless of the parameterization, and that the underlying deformations are quite similar, with the energy only slightly higher in the SVF case. In conclusion, we think the advantages (theoretically guaranteed diffeomorphism with an easy calculation of its inverse) outweigh the disadvantages (possibly lower expressivity, slightly higher final energies in Figure 13), which do not seem to be significant issues for the images and deformations we considered in our experiments.

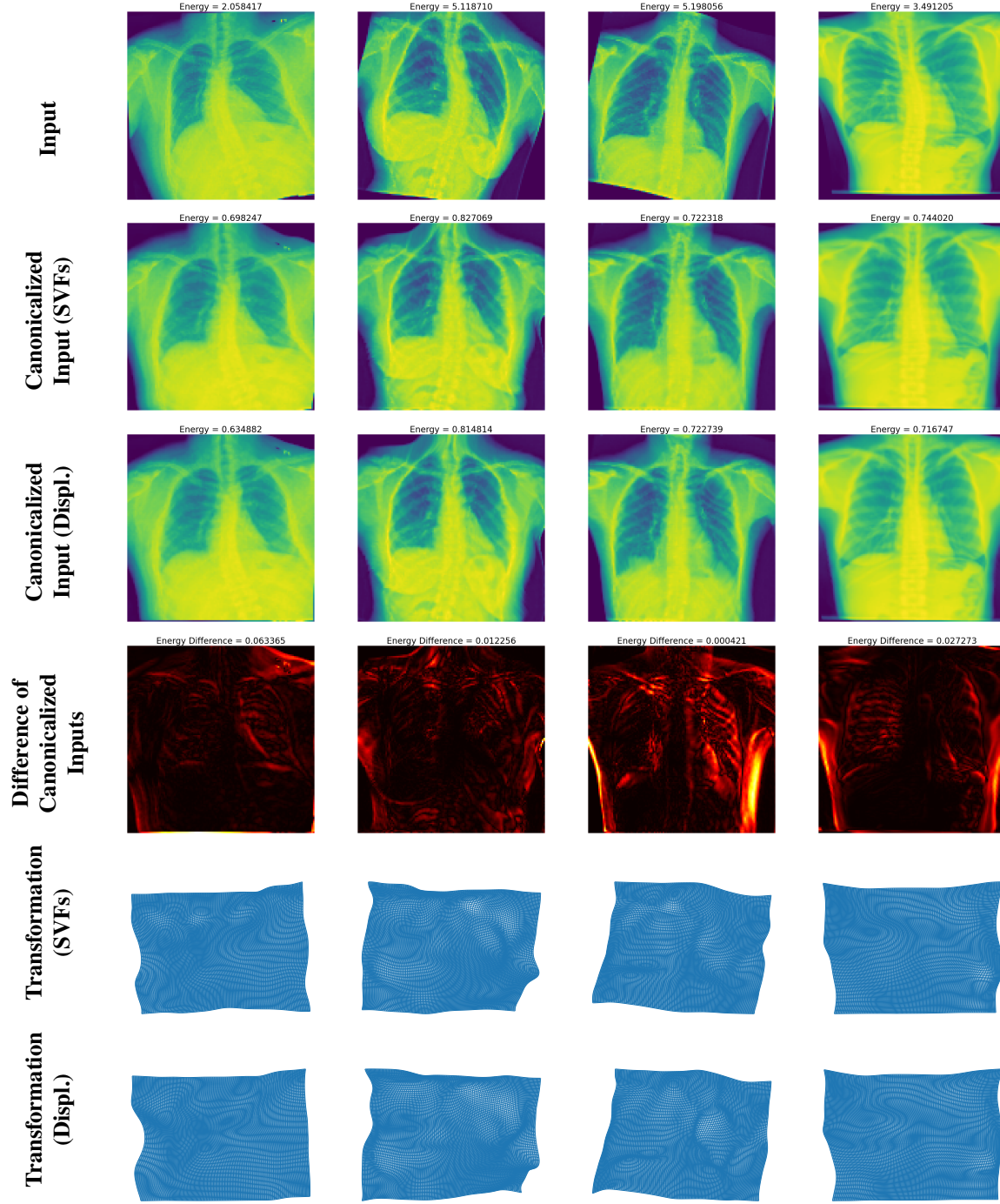


Figure 13. Impact of the parameterization via SVFs in comparison to directly predicting the displacement fields (Displ.). The first row depicts the transformed input images, while the canonicalized versions are shown below using SVFs (second row) and displacement fields (third row), and the difference in the canonicalized versions as well (fourth row). The high similarity of the images indicates that SVFs are sufficiently expressive for this class of transformations. Closer inspection of the deformed grids also shows a high degree of similarity between the generated transformations (last two rows).

D. Canonicalization Energy: Details on the Energy Terms and Ablation Study

The canonicalization energy E_{can} measures the similarity of a transformed image with the training data and encourages more “realistic” transformations within the allowed set of transformations $\mathcal{D}_{\text{SVF}}(\Omega)$. In this work, the energy function $E_{\text{can}} : X \times \mathcal{D}_{\text{SVF}}(\Omega) \rightarrow \mathbb{R}$ consists of a weighted sum of different loss terms:

$$E_{\text{can}}(x, g) := \underbrace{E_{X_E}(g \cdot x)}_{\text{image similarity}} + \underbrace{E_{\text{reg}}(g)}_{\text{regularization}}, \quad (15)$$

where the image similarity energy is defined as

$$E_{X_E}(g \cdot x) := \lambda_{\text{VAE}} E_{\text{VAE}}(g \cdot x) + \lambda_{\text{adv}} E_{\text{adv}}(g \cdot x) \quad (16)$$

and the regularizing energy (following Han et al. (2023) and Bostelmann et al. (2026)) as

$$E_{\text{reg}}(g) := \underbrace{\lambda_{\Delta} \|\nabla v\|_2^2}_{\text{gradient loss}} + \underbrace{\lambda_{\mathcal{J}} \|\max(0, -\det(\mathcal{J}_g))\|_1}_{\text{Jacobian determinant loss}}. \quad (17)$$

Here, λ_{VAE} , λ_{adv} , λ_{Δ} , and $\lambda_{\mathcal{J}}$ are positive scalar weights. The energy functions E_{VAE} (VAE-based energy) and E_{adv} (adversarial energy) evaluate the similarity of an input to the training dataset X_E . The regularizing energy E_{reg} penalizes implausible or undesired transformations.

In the following, we discuss the individual components of our energy in more detail and present an ablation study on the different components, thereby further strengthening our final chosen combination of energy terms.

VAE-based Energy E_{VAE} One of the central parts of E_{can} is the energy function E_{VAE} derived from a *variational autoencoder* (VAE) (Kingma & Welling, 2022). We train the VAE only on the training dataset X_E with the following loss:

Definition D.1 (VAE Loss/ELBO). We consider a VAE with a latent space of dimension $d < n$. For each input $x \in [0, 1]^n$, we assume a Gaussian latent space distribution $q(\cdot|x) \sim \mathcal{N}(\mu, \text{diag}(\sigma^2))$, parameterized by $\mu := (\mu_j)_{j=1}^d \in \mathbb{R}^d$ and $\sigma^2 := (\sigma_j^2)_{j=1}^d \in \mathbb{R}^d$. We define the VAE loss as a combination of *binary cross-entropy* (BCE) and *Kullback–Leibler divergence* (D_{KL} , also called KL divergence) between the encoder distribution and the standard normal distribution:

$$\mathcal{L}_{\text{VAE}}(x) := \underbrace{\text{BCE}(x, \hat{x})}_{\text{reconstruction loss}} + \underbrace{D_{\text{KL}}(q(\cdot|x) \parallel \mathcal{N}(0, I))}_{\text{KL divergence}}, \quad (18)$$

where $\hat{x} \in [0, 1]^n$ is the output of the VAE. Here, the binary cross-entropy is defined as

$$\text{BCE}(x, \hat{x}) := - \sum_{i=1}^n [x_i \ln \hat{x}_i + (1 - x_i) \ln(1 - \hat{x}_i)] \quad (19)$$

with $0 \ln(0) := 0$, and the KL divergence is

$$D_{\text{KL}}(q(\cdot|x) \parallel \mathcal{N}(0, I)) := -\frac{1}{2} \sum_{j=1}^d (1 + \ln \sigma_j^2 - \mu_j^2 - \sigma_j^2). \quad (20)$$

The latter helps to bring the approximate posterior close to a standard normal distribution, while the binary cross-entropy helps to minimize the difference between the input x and the reconstructed input $\hat{x}$. The VAE loss is also called *evidence lower bound* (ELBO). Classically, the first term is more complicated than the BCE. However, as our inputs are normalized to $[0, 1]^n$, we can use the BCE.

We then use the trained VAE to compute a scalar energy value for new inputs. The energy E_{VAE} corresponds to the VAE loss and indicates the distance from an input to the training dataset X_E , on which we train the VAE. Hence, we define the VAE energy $E_{\text{VAE}} : \mathbb{R}^n \rightarrow [0, \infty)$ as

$$E_{\text{VAE}}(x) := \mathcal{L}_{\text{VAE}}(x). \quad (21)$$

A low energy means that the input can be reconstructed well by the VAE, indicating that the input is “close” to the training dataset X_E .

Adversarial Energy E_{adv} To further increase the similarity between a transformed image and the training data, we additionally use an adversarial energy function. The energy function is based on an adversarial discriminator learned by a *discriminator network*, trained to distinguish between the training images X_E and diffeomorphically transformed training images.

The adversarial discriminator is determined by a *Wasserstein generative adversarial network (WGAN)* with a *gradient penalty (WGAN-GP)* framework (Arjovsky et al., 2017). We base our implementation on adversarial regularization (Lunz et al., 2019; Mukherjee et al., 2024), as in (Shumaylov et al., 2024). A discriminator $D : X \rightarrow \mathbb{R}$ is trained in a way that

- $D(x)$ is small if x is “close” to the training data X_E (“real” images), and
- $D(x)$ is large if x is different from the training data X_E (“fake” images).

We generate the “fake” images by applying synthetic transformations to “real” images. This setup defines two implicit data distributions, p_{real} and p_{fake} . These correspond to the dataset X_E and to the distribution of randomly transformed images, and are accessed only via sampling rather than by explicit probability densities.

Let x be an image of the training data X_E and $\hat{x} = g \cdot x$ a generated “fake” image. Furthermore, let the image

$$\tilde{x} = \alpha x + (1 - \alpha)\hat{x}, \quad \alpha \sim \text{Uniform}(0, 1) \quad (22)$$

be a linear interpolation between x and $\hat{x}$, where $\tilde{p}$ denotes the implicit distribution of interpolated samples $\tilde{x}$ between real and fake images. The WGAN-GP loss for training the discriminator D is defined as

$$\mathcal{L}_D(x, \hat{x}) := \mathbb{E}_{\hat{x} \sim p_{\text{fake}}} [D(\hat{x})] - \mathbb{E}_{x \sim p_{\text{real}}} [D(x)] + \mu \cdot \mathbb{E}_{\tilde{x} \sim \tilde{p}} \left[(|\nabla_{\tilde{x}} D(\tilde{x})|_2 - 1)^2 \right], \quad (23)$$

where $\mu \in \mathbb{R}$ is the gradient penalty weight. The term $\mathbb{E}_{x \sim p_{\text{real}}} [D(x)]$ denotes the expectation of the discriminator output over data samples,

$$\mathbb{E}_{x \sim p_{\text{real}}} [D(x)] = \int D(x) dp_{\text{real}}(x). \quad (24)$$

In practice, we approximate the expectation through averaging over the batch size B ,

$$\mathbb{E}_{x \sim p_{\text{real}}} [D(x)] \approx \frac{1}{B} \sum_{i=1}^B D(x_i), \quad x_i \sim p_{\text{real}}(x). \quad (25)$$

We define the expectation analogously for $\hat{x}$ and $\tilde{x}$ with their respective distributions p_{fake} and $\tilde{p}$. The trained discriminator can be used directly to define the adversarial loss:

$$E_{\text{adv}}(x) := D(x). \quad (26)$$

The “closer” the input x is to the training dataset X_E , the smaller the energy $E_{\text{adv}}(x)$. For more details, see (Lunz et al., 2019; Mukherjee et al., 2024; Shumaylov et al., 2024).

One disadvantage of adding the adversarial energy term E_{adv} to the canonicalization energy is that by training the adversarial discriminator for the energy term, we rely on limited data augmentation. However, this is outweighed by two major advantages. Firstly, training an adversarial regularizer requires a much smaller memory and computational footprint than classical augmentation. In our experiments on the chest X-ray dataset, training the adversarial discriminator with approximately 35k weights for the canonicalization energy takes about 10 minutes, whereas training the augmented U-Net with approximately 23.7M weights takes about 50 minutes. Secondly, the adversarial discriminator is trained only on unlabeled image data and is not specific to the task of interest. As such, it is usable for various tasks without retraining. In addition, the ablation study on the energy terms shows that we do not necessarily need the adversarial energy term, but it improves the performance of our network; see Table 2.

Regularizing Energy E_{reg} The proposed regularizing energy E_{reg} (17) is based on the regularization used in (Han et al., 2023; Bostelmann et al., 2026). We penalize deformations that are not smooth or have negative Jacobian determinants by a weighted sum of the gradient loss

$$\|\nabla v\|_2^2 \quad (27)$$

MODEL	IoU $\uparrow$
DIFFEONN WITH $E_{\text{CAN}} = \lambda_{\text{VAE}} E_{\text{VAE}} + \lambda_{\text{ADV}} E_{\text{ADV}} + E_{\text{REG}}$	0.9233
DIFFEONN WITH $E_{\text{CAN}} = \lambda_{\text{VAE}} E_{\text{VAE}} + \lambda_{\text{ADV}} E_{\text{ADV}}$	0.9190
DIFFEONN WITH $E_{\text{CAN}} = \lambda_{\text{VAE}} E_{\text{VAE}}$	0.9179
DIFFEONN WITH $E_{\text{CAN}} = \lambda_{\text{ADV}} E_{\text{ADV}}$	0.6166
DIFFEONN WITH $E_{\text{CAN}} = \lambda_{\text{VAE}} E_{\text{VAE}} + E_{\text{REG}}$	0.9163
DIFFEONN WITH $E_{\text{CAN}} = \lambda_{\text{ADV}} E_{\text{ADV}} + E_{\text{REG}}$	0.9020
NAÏVE	0.9167
AUGMENTED	0.9261

Table 2. Mean Intersection-over-Union (IoU) on 100 randomly chosen and transformed chest X-ray images from (Karmakar, 2024).

and the Jacobian determinant loss

$$\|\max(0, -\det(\mathcal{J}_g))\|_1. \quad (28)$$

The parameter v is the SVF that induces g , and $\det(\mathcal{J}_g)$ denotes the Jacobian determinant of g .

This energy is only one option to regularize, and it could be adjusted task-specifically. Adding a regularization energy term breaks strict equivariance but can be understood as a selection operator. This is discussed in detail in Section 3.2 and Appendix E.

D.1. Ablation Study

To explore the impact of the different energy terms of the canonicalization energy E_{can} (9), we conduct a simple ablation study on 100 randomly chosen and transformed chest X-ray images from the test dataset of (Karmakar, 2024), see also Section 4.1.2. We only vary the energy terms, and otherwise use the implementation of the benchmarking experiments, see Appendix B.1.

We compare the mean Intersection-over-Union (IoU) of the segmentation output and the ground-truth segmentation when using DiffeoNN with different combinations of energy terms. The performances of the naïve and the augmented approach serve as a baseline. The results are presented in Table 2. While the VAE-based energy term E_{VAE} with and without the adversarial energy term E_{adv} both outperform the naïve approach, the combination of all energy terms (including the regularization term E_{reg}) yields the best results. The adversarial energy term E_{adv} , either alone or with the regularization energy term E_{reg} , does not produce sufficient results.

Although the individual energy terms exhibit certain limitations (see discussion above), we consider these trade-offs justified, as their combination yields the most promising results. Consequently, the combination of adversarial, VAE-based, and regularization energy terms emerges as a suitable choice for the canonicalization energy.

E. Regularization as a Selection Operator

In this section, we will show that certain types of regularization act as selection operators. Given such results hold in generality and are largely standard in the inverse problem literature (Benning & Burger, 2018), this should mostly be seen as a recap. We consider the following setting: Let G be a first-countable, Hausdorff topological space (such as a Lie group acting on a space X). We consider the unregularized objective function $f : G \rightarrow \mathbb{R} \cup \{+\infty\}$ defined by $f(g) = E(g^{-1}x)$. We introduce a Tikhonov-style regularization term $\lambda P(g)$ for $\lambda > 0$, where $P : G \rightarrow \mathbb{R}_{\geq 0}$ is a continuous penalty function.

Assumption E.1 (Lower Bounded, Proper and LSC). The function f is proper, bounded below with $m_0 = \inf_{g \in G} f(g) > -\infty$. Furthermore, f is lower semi-continuous (LSC) on G .

Assumption E.2 (Coercivity of the Penalty). The penalty function P is continuous and coercive, meaning that for any $C > 0$, the sub-level set $\{g \in G \mid P(g) \leq C\}$ is compact.

Assumption E.3 (Existence of Unregularized Minimizers). The exact infimum of the unregularized problem is attained. That is, the set of exact minimizers $\mathcal{M} = \{g \in G \mid f(g) = m_0\}$ is non-empty.

Before proving the main selection theorem, we demonstrate that the required minimizers are guaranteed to exist as a direct consequence of coercivity and lower semi-continuity.

Lemma E.4 (Guaranteed Existence of Minimizers). *Under Assumptions E.1 to E.3:*

- (i) For any $\lambda > 0$, the regularized objective $F_\lambda(g) = f(g) + \lambda P(g)$ admits a global minimizer $g_\lambda \in G$.
- (ii) The set of penalty-minimizing exact solutions, $\mathcal{M}_P = \arg \min_{g \in \mathcal{M}} P(g)$, is non-empty.

Proof. Firstly, we establish the existence of g_λ . Choose any $g_0 \in \text{dom}(f)$ (where $f(g_0) < \infty$) and let $C = F_\lambda(g_0)$. The specific sub-level set $\{g \mid F_\lambda(g) \leq C\}$ is nonempty (as it contains g_0) and is contained within $\{g \mid P(g) \leq (C - m_0)/\lambda\}$. By Assumption E.2, the latter is compact. Since f is LSC and P is continuous, F_λ is LSC. The sub-level sets of F_λ are therefore closed subsets of a compact set, making them compact. An LSC function on a compact space achieves its minimum, guaranteeing the existence of g_λ .

Secondly, we establish the existence of $\mathcal{M}_P$. By Assumption E.3, $\mathcal{M}$ is non-empty. Because f is LSC, $\mathcal{M} = \{g \in G \mid f(g) \leq m_0\}$ is a closed set. Let $g_0 \in \mathcal{M}$ be an arbitrary exact minimizer. To minimize P over $\mathcal{M}$, it suffices to search within the restricted set $\mathcal{M} \cap \{g \in G \mid P(g) \leq P(g_0)\}$. This set is the intersection of a closed set and a compact set (due to the coercivity of P), hence it is compact. Because P is continuous, it attains its minimum on this compact set, guaranteeing that $\mathcal{M}_P$ is non-empty. $\square$

With this, we can finally show the main result, establishing selection.

Theorem E.5. *Assume Assumptions E.1 to E.3 hold. As $\lambda \rightarrow 0^+$, the sequence of regularized minimizers $\{g_\lambda\}_{\lambda>0}$ is contained within a compact set. Furthermore, every accumulation point $\bar{g}$ of this sequence as $\lambda \rightarrow 0^+$ belongs to $\mathcal{M}_P$.*

Proof. Let $g^* \in \mathcal{M}_P$ be an arbitrary but fixed penalty-minimizing solution to the unregularized problem (whose existence is guaranteed by Theorem E.4). Since g_λ is a global minimizer of the regularized problem, we have:

$$f(g_\lambda) + \lambda P(g_\lambda) \leq f(g^*) + \lambda P(g^*). \quad (29)$$

By definition, $g^* \in \mathcal{M}$, which implies $f(g^*) = m_0$. Additionally, since m_0 is the global infimum of f , we have $f(g_\lambda) \geq m_0$. Substituting these bounds into (29) yields:

$$m_0 + \lambda P(g_\lambda) \leq m_0 + \lambda P(g^*). \quad (30)$$

Subtracting m_0 and dividing by $\lambda > 0$, we obtain:

$$P(g_\lambda) \leq P(g^*). \quad (31)$$

This establishes that for all $\lambda > 0$, the minimizers g_λ belong to the sub-level set $\{g \in G \mid P(g) \leq P(g^*)\}$. By Assumption E.2, this sub-level set is compact.

Let $\lambda_n \rightarrow 0^+$ be an arbitrary sequence. Because $\{g_{\lambda_n}\}$ lies in a compact set, and G is first-countable, it admits a convergent subsequence. Without loss of generality, let $\{g_{\lambda_n}\}$ denote this convergent subsequence with limit point $\bar{g} \in G$.

We first show that $\bar{g}$ is an exact minimizer of f . Rearranging (29) evaluated at λ_n , we obtain an upper bound for $f(g_{\lambda_n})$:

$$f(g_{\lambda_n}) \leq m_0 + \lambda_n (P(g^*) - P(g_{\lambda_n})). \quad (32)$$

Taking the limit inferior as $n \rightarrow \infty$, and exploiting the lower semi-continuity of f , we have:

$$f(\bar{g}) \leq \liminf_{n \rightarrow \infty} f(g_{\lambda_n}) \leq \lim_{n \rightarrow \infty} [m_0 + \lambda_n (P(g^*) - P(g_{\lambda_n}))]. \quad (33)$$

Because $P(g_{\lambda_n}) \leq P(g^*)$ for all n , the sequence of real numbers $P(g_{\lambda_n})$ is bounded. Thus, the term multiplied by λ_n vanishes as $\lambda_n \rightarrow 0^+$. Hence,

$$f(\bar{g}) \leq m_0. \quad (34)$$

Since m_0 is the global infimum, it must be that $f(\bar{g}) = m_0$, which implies $\bar{g} \in \mathcal{M}$.

Finally, we show that $\bar{g}$ minimizes the penalty within $\mathcal{M}$. By the continuity of P and the inequality established in (31):

$$P(\bar{g}) = \lim_{n \rightarrow \infty} P(g_{\lambda_n}) \leq P(g^*). \quad (35)$$

Because $\bar{g} \in \mathcal{M}$ and g^* is defined as an element that yields the minimum possible penalty in $\mathcal{M}$, the inequality $P(\bar{g}) \leq P(g^*)$ forces $P(\bar{g}) = P(g^*)$. Therefore, $\bar{g} \in \mathcal{M}_P$.

This concludes the proof that any accumulation point of $\{g_\lambda\}$ is a penalty-minimizing exact solution. $\square$

F. Homology Classification on MNIST

F.1. Setup and Implementation Details

Dataset The dataset is generated from the MNIST handwritten digit dataset (LeCun et al., 1998) by constructing a topology-preserving classification problem based on the number of enclosed holes in each digit. First, digits $\{2, 4\}$ are excluded, and the remaining digits are assigned to one of three classes according to their hole count:

$$\text{Class 0 : } \{1, 3, 5, 7\}, \quad \text{Class 1 : } \{0, 6, 9\}, \quad \text{Class 2 : } \{8\}.$$

To avoid class imbalance, a balanced subset is created by randomly sampling an equal number of images from each hole-count group, corresponding to the minimum available class size. The dataset is then partitioned into 2000 training, 500 validation and 500 test samples.

For every selected image, we additionally generate a transformed version through a smooth random diffeomorphic deformation, similar to the other datasets, see Appendix A.1 and B.1. Specifically, a regular grid of 16 control points (4×4) is initialized within the normalized image coordinate system $[-0.5, 0.5]^2$ and scaled inward by a factor of 0.6. A random affine transformation is then applied, consisting of a rotation $\theta \sim \mathcal{U}(-\frac{\pi}{3}, \frac{\pi}{3})$, combined with a scaling factor of 1.5. In addition, each control point receives an independent local displacement sampled uniformly with a displacement vector with maximum norm 0.2. Additionally, all points are clipped to remain inside the image domain.

The sparse deformation field is interpolated into a dense displacement vector field (DVF) using a thin-plate spline radial basis function (RBF) interpolator. The resulting dense deformation is evaluated on the full image grid and applied using a bilinear spatial transformer. Finally, transformed images are intensity-clamped to the range $[0, 1]$ and saved together with their corresponding topology label (number of holes).

Homology Classification Network For the inner network, a convolutional neural network (CNN) was implemented to estimate the number of holes present in a digit image, serving as an auxiliary structural feature extractor. The model accepts a grayscale input image and consists of two convolutional blocks. The first block applies 16 convolutional filters of size 3×3 , followed by a Rectified Linear Unit (ReLU) activation and 2×2 max pooling. The second block increases the feature depth to 32 filters using the same convolutional configuration, again followed by ReLU activation and max pooling. The resulting feature representation is flattened and passed through a fully connected classifier composed of a hidden layer with 64 neurons and a final output layer with three logits corresponding to the possible hole counts/homology classifications (0, 1, or 2).

The inner network was trained using the Adam optimizer (Kingma & Ba, 2017) with a learning rate of 10^{-3} and optimized using cross-entropy loss. Training was conducted for 10 epochs with a batch size of 64, using shuffled mini-batches from the training set and a separate validation split for model selection. The model checkpoint achieving the highest validation accuracy was retained for evaluation on the test set.

The augmented network was trained in the same manner. Only the training dataset was adjusted: the training dataset contained the original training dataset of the inner network and additionally one diffeomorphically transformed version of each of the training samples.

Canonicalization Step Except for slight dimensional changes, the canonicalization energy was trained as in Appendix A.1 and B.1 on the original dataset. The latent space of the VAE was adjusted to dimension 10.

In the canonicalization step, we use 100 gradient steps to find the canonicalization and set $\lambda_{\text{adv}} = 0.01$, $\lambda_{\text{VAE}} = 0.01$, $\lambda_{\Delta} = 1$, and $\lambda_{\mathcal{J}} = 10$.

The detailed source code is available at <https://github.com/mic-uzl/DiffeoNN>.

F.2. Visual Results

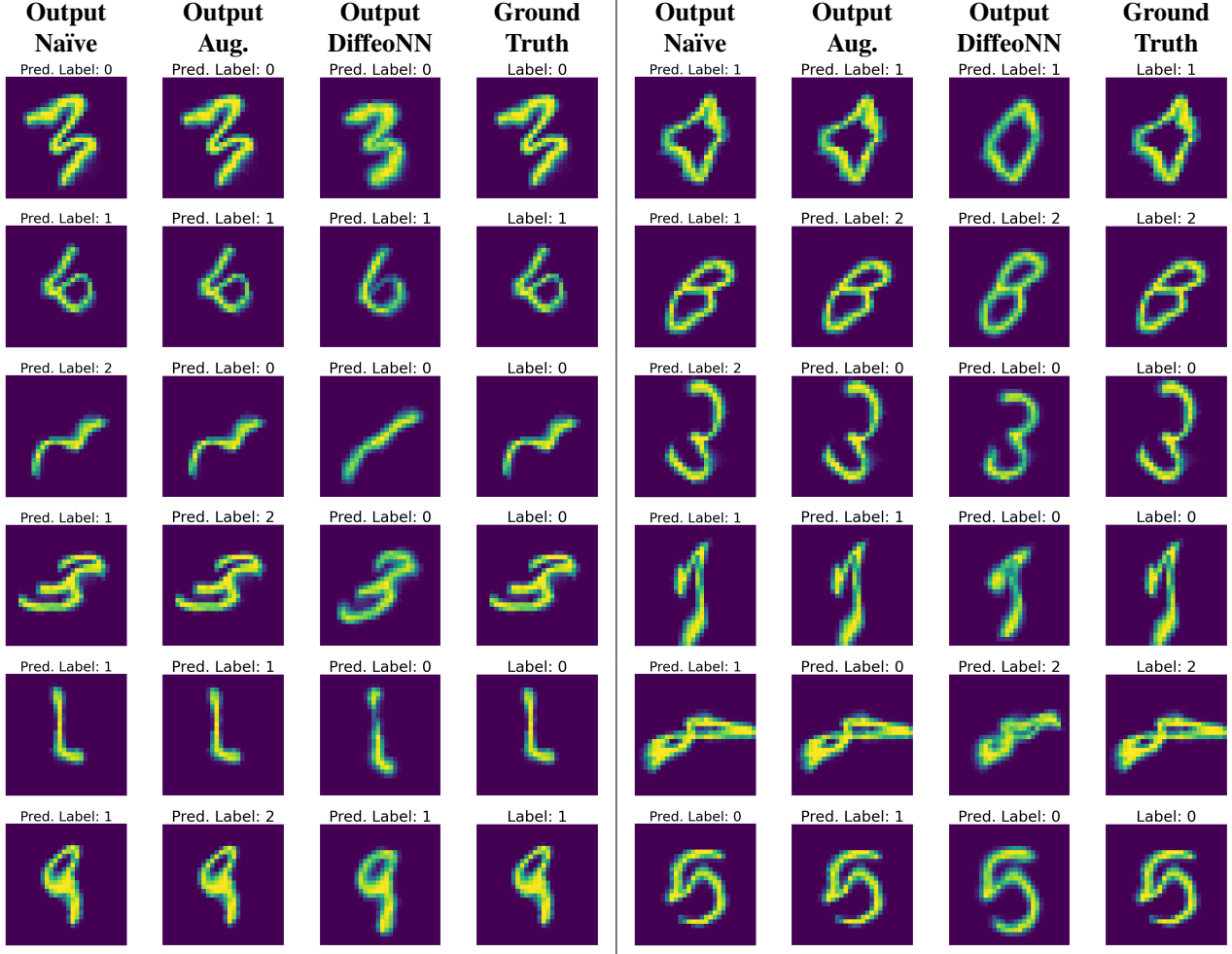


Figure 14. Homology classification task on 100 diffeomorphically transformed MNIST (LeCun et al., 1998). Shown are the input image with the predicted genus by the naïve approach (**column one**), the augmented approach (**column two**), and the ground truth (**column four**) as well as the canonicalized input and the prediction by DiffeoNN (**column three**). For DiffeoNN and the naïve approach, we train a simple convolutional classifier for determining the genus, an adversarial network and a VAE with latent space dimension 10 on the original MNIST. The augmented classifier is trained similarly but on a dataset containing original and transformed MNIST images. The canonicalized images in DiffeoNN are visually “closer” to the original MNIST. The naïve classifier struggles most with the transformed input images, but also the augmented approach predicts incorrect labels, see last three rows.

G. 3D Lung Segmentation: Preliminary Visual Results

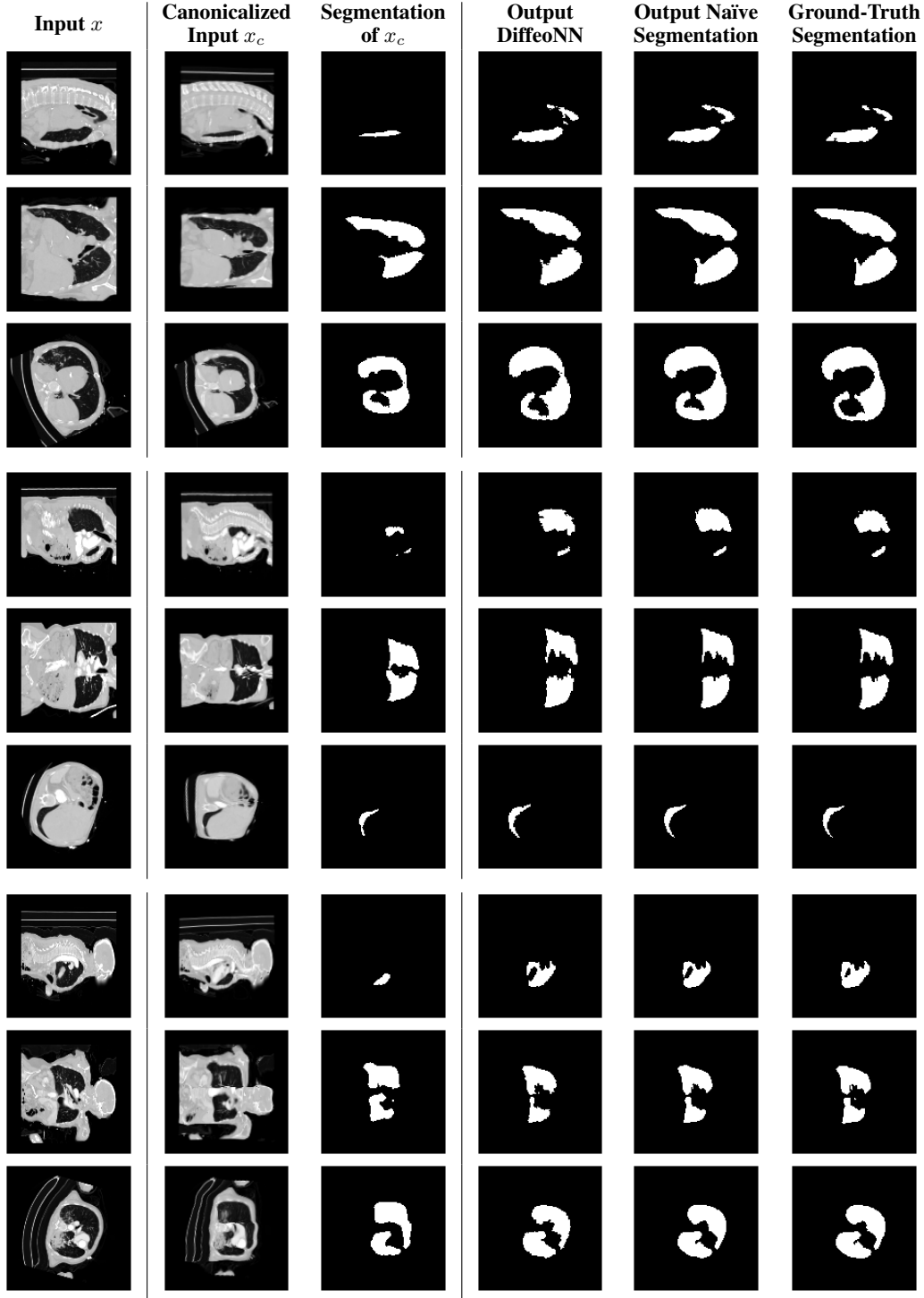


Figure 15. Lung segmentation on three-dimensional CT data (Wasserthal, 2023). We use the pre-trained TotalSegmentator (Wasserthal et al., 2023) as an inner network and compare DiffeoNN with the naïve approach. Groups of three rows present one example, showing one slice of each of the three axes. The canonicalization approach is successful, but the canonicalized images do occasionally have warping artifacts (see wavy torso in the last two rows). The segmentation outputs of DiffeoNN and the naïve TotalSegmentator are close to the ground truth segmentation. In example one (rows one to three), DiffeoNN produces a more accurate segmentation than the naïve approach.